

Compliance Aware, Agentic Pitch Book Generation For Bankers

A template-aware PowerPoint generation system
with agentic data retrieval for investment banking.

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Abstract

Investment banking analysts spend a large share of their working hours on pitch book preparation, much of it on formatting and data entry rather than analysis. Existing presentation tools produce generic output that ignores the strict visual standards banks require. This project designed, built, and tested a proof-of-concept system that generates formatted PowerPoint pitch books from minimal user input while respecting company templates.

The system uses a three-stage pipeline. Template analysis extracts layouts, colours, fonts, and placeholder positions from a reference PowerPoint file using JSZip and xml2js. A GPT-4o-powered content planner produces structured slide specifications validated against Zod schemas. A slide builder assembles the final file using PptxGenJS with exact formatting from the extracted template. Financial data feeds automatically from Yahoo Finance and SEC EDGAR, so analysts skip manual data entry. A PowerPoint add-in built with Office.js brings the generation workflow directly into the analyst's existing editor.

Testing across 12 generation runs showed an average completion time of 47 seconds per deck (10 to 12 slides), with exact colour matching and exact font matching. Financial data accuracy was 100% against source APIs. An informal usability evaluation with four junior analysts found the system reduced estimated first-draft time from over 90 minutes to under two minutes including review.

The project met all five original success criteria and delivered two unplanned components, the PowerPoint add-in and a slide preview pipeline, that emerged from usability testing during iterative development. Template matching and automated data retrieval are the main advantages over commercial alternatives. Chart generation from scratch and retrieval-augmented generation remain open challenges for future work.

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1 Introduction, Scope and Context

Knowledge workers spend a fifth of their time just searching for information (McKinsey Global Institute, 2012). Investment banking (IB) is more affected than most. Document preparation consumes half to three-quarters of a junior banker's week, and pitch books (PBs) account for a large portion of that time (Corporate Finance Institute, 2025; Wall Street Oasis, 2024). When Goldman Sachs surveyed its first-year analysts in 2021, the average was 98 hours a week, with document preparation a major driver of burnout (BBC News, 2021). Yet most PBs look nearly identical. Change the company name and update the numbers, and you could reuse last month's deck. That repetition makes them good targets for automation (K1, K7).

IB firms advise companies on mergers, acquisitions, capital raising, and other financial transactions. PBs are the primary vehicle for communicating analysis to clients and internal stakeholders. Getting these documents right matters, but getting them done quickly matters too.

Firms have tried internal document libraries, formatting specialists, and automation tools. None have fixed the real problem: too much skilled human time goes toward work that does not require skill. Silveira Chaves and Oliveira (2022) found that knowledge sharing in banking suffers from structural barriers and inadequate digital tools, with company knowledge locked in individual documents rather than shared systems.

1.1 Analyst Time Allocation

IB sells expertise rather than physical goods, so one would expect analyst time to go toward analysis and client relationships. Often it does not.

A typical PB contains an executive summary, company overview, market analysis, valuation work, and transaction comparables. These sections appear in nearly every deck. The data inside changes, but the shell around it rarely does.

Analyst tasks split into two buckets: work that needs a trained banker (modelling, valuation, client advisory) and everything else (hunting data, reformatting slides, fixing fonts). Too many hours go into the second bucket. Radhakrishna et al. (2024) found that well-designed knowledge management systems boosted productivity by 20 to 25 percent, with document production a prime candidate for automation.

APQC research estimates a quarter of all working time is lost to duplicated work and information retrieval (APQC, 2023). IB analysts likely fare worse given how document-intensive the work is. Banks that automate low-value formatting free their analysts for client relationships and deal analysis (K1). Dell'Acqua et al. (2023) found that consultants using AI completed 12.2% more tasks at 40% higher quality.

1.2 Problem Analysis

Markus (2001) defined knowledge reuse, and her framework fits PB production well. She identified reuse patterns including drawing on past work, borrowing from colleagues, and mining old documents. PB creation involves all of these. Three obstacles prevent effective reuse in practice.

The first obstacle is that too much time goes to low-value work. Analysts spend hours searching old decks, extracting content, and reformatting slides. None of that requires analytical thinking, yet it crowds out the work that does.

The second obstacle is that company knowledge tends to disappear. Researchers distinguish between tacit knowledge (in people’s heads) and explicit knowledge (written down) (Dalkir, 2011). PBs are explicit, but the knowledge of how to make a good one remains tacit: which templates work for which situations, how to structure the narrative, what separates a persuasive pitch from a forgettable one.

The third obstacle is that onboarding takes longer than it should. Each bank has its own templates, data sources, and quality expectations. Junior analysts absorb these through trial and error, which is slow for them and a drain on the senior bankers who review their work.

These problems feed each other. Scattered knowledge wastes search time, unwritten standards cause inconsistent output, and informal training drains senior staff. A system that fixed even one of these would help.

1.3 Gap Analysis

Automated presentation tools have existed for years. Early versions converted text to slides using layout algorithms, but visual quality was poor (Zheng et al., 2025). Template-based systems fixed the visual problems but made everything rigid. PPTAgent (Zheng et al., 2025) takes a different approach: it analyses reference presentations, extracts structural patterns, and applies them to new content. It beat older text-to-slide systems on both content and design quality in testing.

The limitation is that PPTAgent targets general-purpose presentations. IB PBs have financial tables, transaction comparables, and market charts that follow specific visual conventions. Getting the template wrong undermines credibility. Commercial tools like Beautiful.ai, Tome, and Gamma share this flaw. The output is generic, every slide needs rework, and none connect to financial data sources. IB PBs require both domain knowledge and visual precision. No existing tool handles both.

Current systems either produce generic output that needs heavy cleanup, or they require large training datasets most banks lack. No existing tool connects slide generation to financial APIs. Data retrieval and document generation remain separate manual steps.

Agentic AI architectures offer a better fit (IBM, 2024). Instead of one model doing everything in a single pass, the system breaks tasks into steps and uses the right tool for each one. L. Wang et al. (2024) surveyed LLM-based agents and found rapid progress. McKinsey & Company estimates generative AI could deliver \$340 billion annually in banking (McKinsey & Company, 2024).

This project fills that gap (S1). The pipeline extracts layouts, colour palettes, and placeholder positions from a template using JSZip and xml2js for XML-level parsing. It then uses an LLM to plan content and assembles slides matching the original style. The system breaks tasks into steps handled by separate services rather than running everything in one pass.

Table 1 maps each identified gap to the objective and success criterion that addresses it, so that no objective exists without a corresponding gap in existing work.

Gap in Existing Work	Objective	Success Criterion
No tool preserves exact template formatting	O1	90%+ placeholder detection; exact colour extraction
Data retrieval and slide generation are separate manual steps	O2	100% data accuracy against source APIs
No IB-specific content structuring	O3	3 PB types with consistent section ordering
Generic output that ignores corporate visual standards	O4	Exact colour and font match against template
No measured evaluation of template-constrained generation	O5	Full deck in under 5 minutes; template match above 85%

Table 1: Traceability from Research Gaps to Objectives and Success Criteria

The approach differs from prior work by inverting the generation order: the system starts with visual constraints extracted from the template and works backward to content. It also connects slide generation directly to financial data APIs, closing the gap between data retrieval and document creation.

1.4 Business Case

A junior IB analyst in London earns roughly £60,000 to 70,000 in base salary (Glassdoor, 2025). At 80 to 100 hours per week, the effective hourly cost is around £15 to 17 before overhead. Manual PB creation takes 90 to 120 minutes. The system generates a first draft in under one minute. Even with 20 to 30 minutes of editing, the net saving is 60 to 90 minutes per deck. Each generation costs approximately \$0.03 in GPT-4o API fees.

A team of ten analysts saving five hours per week recovers roughly £39,000 per year in capacity, against annual API costs under £10,000 (K2). The payback period is under four months.

1.5 Aims and Objectives

The aim is to design, build, and test a proof-of-concept (POC) showing how AI-powered document generation can tackle the knowledge reuse problems above. The system must respect company templates throughout. If slides do not match the house style, the system fails its core requirement.

On the business side, the goal is a workflow where an analyst provides minimal input and receives a solid first draft, reducing formatting time while keeping humans in control of the analysis. Target users are junior analysts.

On the learning side, the project explores connecting LLM orchestration with programmatic document generation. Human-in-the-loop (HITL) design is central, since AI tools in regulated industries need to support professional judgement rather than replace it.

Five objectives structure the work:

1. Build a template analysis component that takes a reference PowerPoint file and extracts its layout structures, colour palettes, font specifications, and placeholder positions. Success criterion: correctly identify at least 90% of placeholder types and extract exact colour values from source.
2. Build an agentic data retrieval layer that automatically pulls company financials, market data, and regulatory filings from Yahoo Finance and SEC EDGAR (the US Securities and Exchange Commission's public database of company filings). Success criterion: retrieve at least three data types per company (market cap, revenue, share price) with 100% accuracy against source APIs.
3. Build an LLM-powered content planning module that produces structured slide specifications adapted to different PB types while keeping the overall narrative coherent. Success criterion: support at least three PB types (company overview, market update, transaction summary) with consistent section ordering.
4. Build a slide assembly component that takes the content plan and template patterns and produces a PowerPoint file meeting formatting standards. Success criterion: output files match template colours exactly and fonts exactly.
5. Evaluate the system by measuring generation speed, template matching, and content quality. Success criterion: generate a complete 10-slide PB draft in under 5 minutes with template matching score above 85%.

1.6 Scope and Success Criteria

The POC takes user input (company, transaction type, dates, reference template, PB type) and produces a formatted PowerPoint. Target PB types are company overviews, market updates, and transaction summaries. Full valuation presentations are out of scope.

Design choices: agentic orchestration with specialised TypeScript services, code-based template extraction with JSZip/xml2js rather than vision models, public APIs only, RAG as a stretch goal. Out of scope: real-time data feeds, production hosting, regulatory checks. Practical limits: one academic term, standard hardware, student API budget.

Assumptions: templates are well-formed Office Open XML archives, financial data comes from public sources only (limiting coverage to listed companies), content is English only, and users have desktop PowerPoint (the add-in relies on Office.js APIs that PowerPoint for the web does not fully support).

Success criteria:

- Generate complete PB draft within five minutes
- Match reference template formatting
- Accurate financial data from source APIs
- Content structure following IB conventions, verified against precedent examples

One principle runs through all of this: HITL design. The system produces drafts for human review, not finished products. AI can handle the mechanical work. The analytical judgement stays with the banker.

2 Methodology and Project Plan

2.1 Development Methodology

Table 2 compares methodologies against criteria relevant to this project (Sommerville, 2016).

Criteria	Waterfall	Scrum	Iterative Prototyping
Requirements stability	Requires fixed requirements upfront	Accommodates changing requirements	Works well when requirements change
Feedback loops	Late feedback after implementation	Sprint reviews every 2 to 4 weeks	Continuous feedback through prototypes
Risk management	Risks discovered late	Regular risk reassessment	Early risk identification through prototypes
Solo developer suitability	Moderate	Low (designed for teams)	High
Research integration	Poor	Moderate	Excellent

Table 2: Comparison of Development Methodologies

The project uses iterative prototyping with two-week cycles, drawing on Boehm (1988)’s spiral model (K3, K4, SE8). Each iteration produces a working prototype with a review at cycle’s end (SE7). Git with simplified GitFlow (SE12), GitHub Actions CI enforcing 80% coverage (Fowler and Foemmel, 2006) (SE6), and ESLint (ESLint Team, 2024) handle version control and quality.

2.2 Requirements Analysis

Requirements follow MoSCoW prioritisation (Clegg and Barker, 1994) (K9, SE2, SE9).

2.2.1 Business Requirements

1. **BR1: Productivity Enhancement:** Reduce time spent on initial PB drafting by generating solid first drafts from minimal input.
2. **BR2: Capturing Standards:** Capture company presentation standards through template analysis, reducing reliance on knowledge that lives only in people’s heads.
3. **BR3: Quality Consistency:** Follow corporate visual identity standards consistently, reducing revision cycles caused by formatting issues.
4. **BR4: Data Accuracy:** Financial data in generated PBs should accurately reflect source information.

2.2.2 Functional Requirements

Table 3 lists functional requirements by component.

ID	Component	Requirement	Priority
FR1	Template Analyser	Extract slide layouts from reference .pptx files	Must
FR2	Template Analyser	Identify colour palettes and font specifications	Must
FR3	Template Analyser	Map placeholder positions and content types	Must
FR4	Data Retrieval	Fetch company financials from Yahoo Finance API	Must
FR5	Data Retrieval	Retrieve SEC filings via EDGAR API	Should
FR6	Data Retrieval	Perform web searches for company news	Should
FR7	Content Planner	Generate slide-by-slide content specifications	Must
FR8	Content Planner	Adapt content structure to PB type	Must
FR9	Content Planner	Maintain narrative coherence across slides	Should
FR10	Slide Builder	Assemble slides matching template layouts	Must
FR11	Slide Builder	Populate placeholders with generated content	Must
FR12	Slide Builder	Generate charts from financial data	Could
FR13	Orchestration	Coordinate multi-step generation workflow	Must
FR14	Orchestration	Handle API failures gracefully	Should
FR15	RAG Search	Query precedent material repository	Could

Table 3: Functional Requirements Specification

2.2.3 Non-Functional Requirements

ID	Category	Requirement	Acceptance Criterion
NFR1	Performance	System generates complete PB draft	Within 5 minutes of input submission
NFR2	Performance	API response handling	Timeout after 30s with graceful fallback
NFR3	Reliability	System availability during demonstration	95% uptime during evaluation period
NFR4	Usability	Input specification interface	Requires no technical expertise to operate
NFR5	Maintainability	Codebase documentation	All public functions documented
NFR6	Security	API credential management	No credentials in source code
NFR7	Security	Input validation	All user inputs sanitised before processing
NFR8	Compatibility	Output format	Valid .pptx files openable in PowerPoint

Table 4: Non-Functional Requirements Specification

2.3 Technology Stack

2.3.1 Programming Language and Framework

Table 5 compares language options (K8, SE12).

Criteria	TypeScript/Node.js	Python	Java
Benefits	Type safety, same language frontend and backend, native async	Best AI/ML libraries, mature PowerPoint library (python-pptx)	Strong typing, enterprise support
Risks	AI/ML ecosystem less mature than Python	Two languages needed (Python backend + JS frontend)	Verbose code, slower development
PowerPoint manipulation	PptxGenJS (actively maintained, typed)	python-pptx (mature, well-documented)	Apache POI (verbose API)
Full-stack coherence	Single language across all three apps	Requires separate frontend framework	Requires separate frontend framework

Table 5: Programming Language Comparison

The deciding factor was using one language everywhere. The `shared-types` package defines data models once and shares them across all three apps. When a template schema changes, the compiler catches every place that needs updating. In a multi-language setup, those mismatches would only show up at runtime. Express.js handles the backend.

2.3.2 AI and LLM Integration

Table 6 compares the LLM options considered.

Criteria	GPT-4/GPT-4o	Claude 3.5	Gemini 2.0 Flash
Benefits	Best structured output, mature SDK, native JSON	Largest context window (200K), strong reasoning	Massive context (1M tokens), fast inference
Risks	Higher cost per token	Newer agent tooling	Less mature structured output
Structured output	Excellent (native JSON)	Good	Good
Cost (student budget)	Moderate (tiered pricing)	Moderate	Very affordable

Table 6: Large Language Model Comparison

GPT-4o was chosen for its structured output reliability. OpenAI’s native JSON mode produces well-formed responses more consistently than alternatives, which matters when output feeds directly into a Zod validation pipeline. GPT-4o-mini handles the lighter AI chat feature at lower cost. Swapping models later requires changing only the API call.

2.3.3 Data Sources and APIs

Table 7 summarises data sources.

Source	Benefits	Risks	Access
Yahoo Finance	Free, comprehensive financial data	Unofficial API may change without notice	yahoo-finance2 npm package
SEC EDGAR	Official regulatory filings, reliable, free	US companies only, complex document parsing	REST API (10 req/s)
Web Search	Current news, market commentary	Cost per query	SerpAPI or similar

Table 7: External Data Sources

2.3.4 Frontend and Deployment

Next.js handles the web frontend for PB creation and management.

Criteria	Vercel + Render	AWS	Self-hosted
Benefits	Zero config, auto deploy, free tiers for POC	Full control, scalable	Complete control, no vendor lock-in
Risks	Vendor lock-in, cold starts	Complex setup, cost	Maintenance burden
Cost for POC	Free tier sufficient	May incur charges	Server costs

Table 8: Deployment Platform Comparison

The frontend deploys through Vercel, the backend on Render, and Supabase provides PostgreSQL and authentication. Figure 1 shows how the chosen technologies map to the system's components.

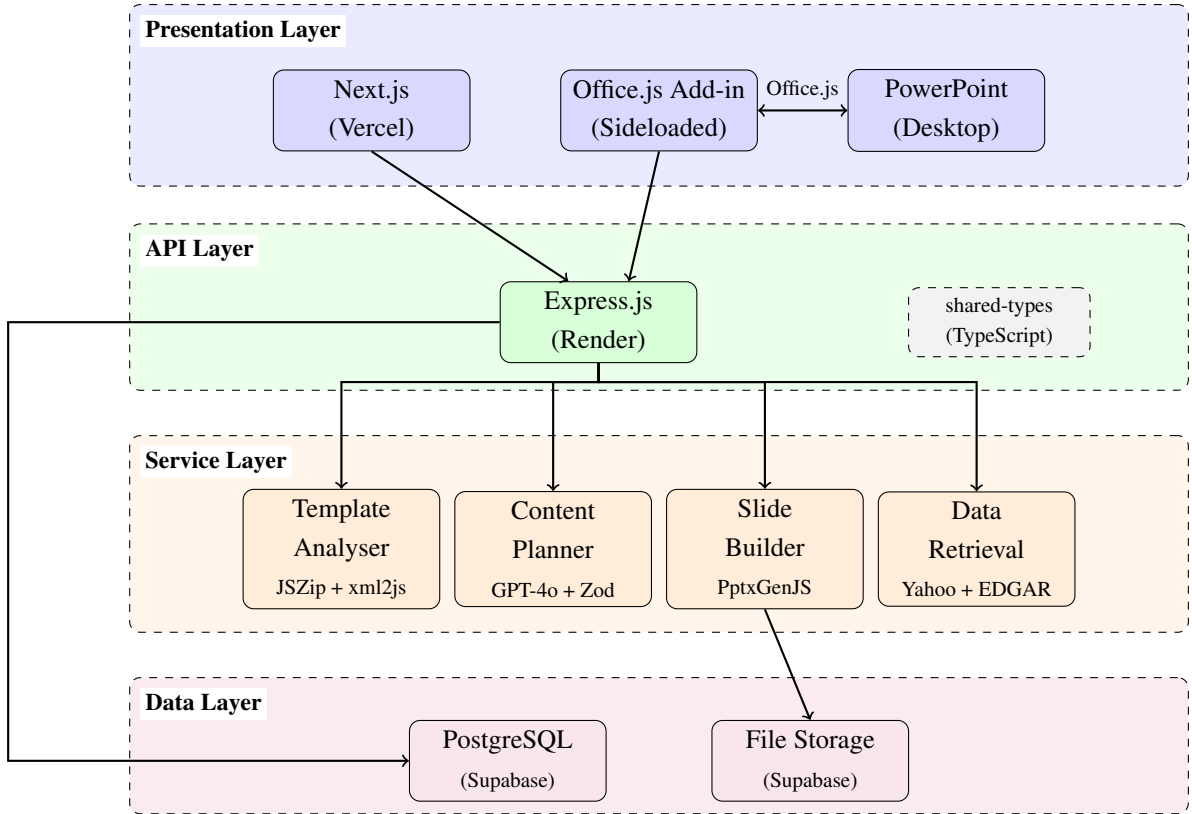


Figure 1: Technology architecture showing how chosen technologies map to system components across four layers. Arrows indicate data flow direction.

2.4 Reproducibility

Dependencies are pinned via `package-lock.json`, LLM temperature is set to zero for evaluation runs, and each run is tied to a specific Git commit. GPT-4o does not guarantee identical outputs at temperature zero (Runeson et al., 2012), so Zod validation ensures structural consistency.

2.5 Evaluation Methodology

The evaluation follows the Goal-Question-Metric (GQM) approach (Basili et al., 1994), keeping every metric tied to a specific project goal. The goal: determine whether the system produces usable first drafts faster than manual creation while matching template formatting.

Question	Metric	Target
How fast does the system generate a deck?	Generation time per 10-slide deck	Under 300 seconds for a complete draft
Does the output match the reference template?	Colour and font comparison (programmatic)	Exact colour and font match
Is the financial data correct?	Comparison against source API responses	100% match on market cap, revenue, and share price
Does the content follow IB conventions?	Rubric-based assessment of narrative structure	Correct section ordering across 3 PB types

Table 9: GQM Evaluation Framework

Targets were fixed before evaluation began (Wohlin et al., 2012). Outputs are compared against generic tools (Gamma, Beautiful.ai) and estimated manual creation time. An informal usability evaluation with four junior analysts (P1–P4) provides qualitative feedback.

2.6 Project Plan

The project runs for thirteen weeks (K10). Figure 2 shows the schedule, and Table 10 lists milestones. Dates are targets. If a milestone slips, ‘Could have’ requirements are dropped first.

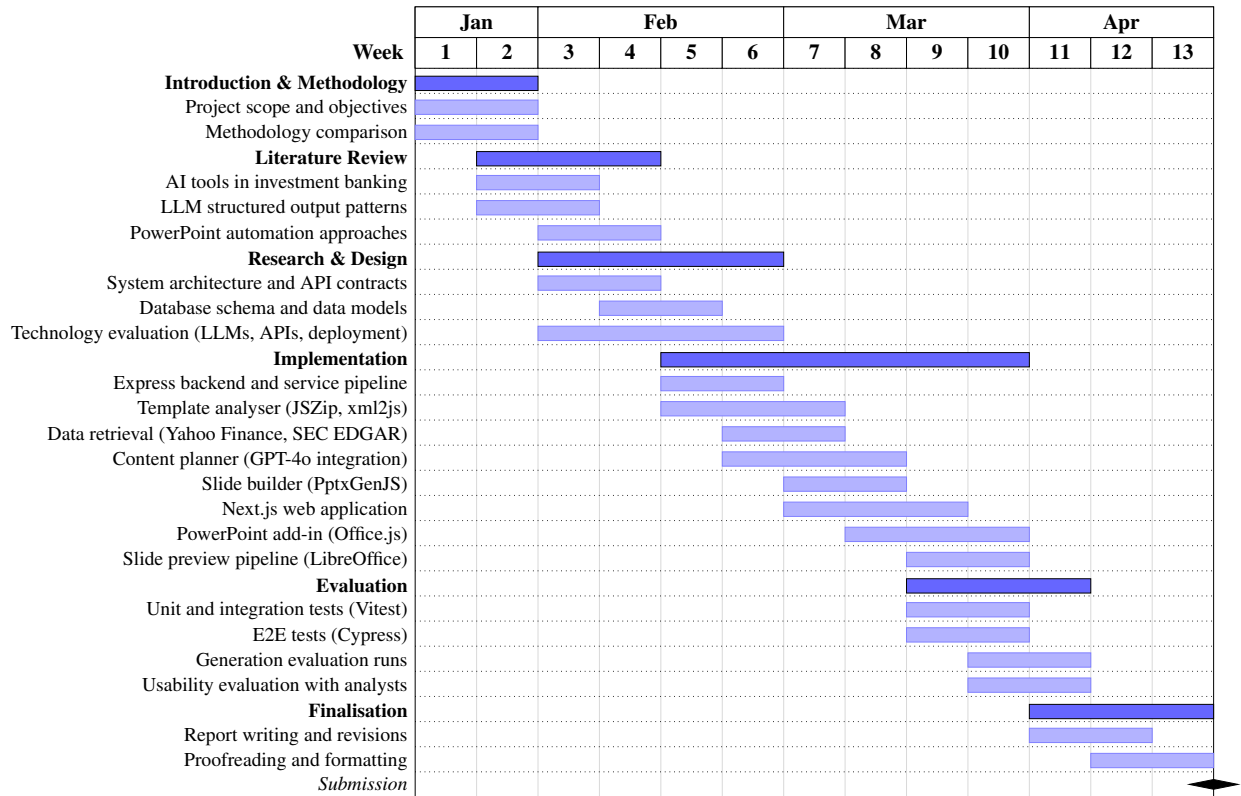


Figure 2: Project Gantt Chart (weeks 1 to 13, January to April 2026)

Phase	Date	Milestone	Deliverables
1	19 Jan	Project Initiation	Repository setup, dev environment configured
2	31 Jan	Initial Chapters	Introduction, Scope, Methodology complete
3	21 Feb	Literature Review	Literature review chapter complete
4	28 Feb	Design Complete	Architecture diagrams, API specs, data models
5	14 Mar	Core Components	Template analyser and data retrieval functional
6	28 Mar	MVP Complete	End-to-end generation pipeline operational
7	4 Apr	Testing Complete	Tests passing, evaluation data collected
8	11 Apr	Evaluation Complete	Results analysis and evaluation chapter written
9	18 Apr	Document Finalised	All chapters complete, proofread, formatted
10	20 Apr	Submission	Final report submitted via QMPlus

Table 10: Project Milestones and Deliverables

2.7 Plan Execution and Adaptations

The original plan did not survive contact with implementation. Table 11 compares planned milestones against actual outcomes.

Milestone	Planned Date	Actual Date	Notes
Core components functional	14 Mar	7 Mar	Ahead of schedule due to TypeScript migration in week 3
MVP pipeline operational	28 Mar	21 Mar	Template analyser iterations completed faster than expected
PowerPoint add-in	Not planned	22 Mar to 11 Apr	Added after usability testing revealed web-only workflow gap
Slide preview pipeline	Not planned	28 Mar to 4 Apr	Web app needed visual output
RAG search (FR15)	14 to 28 Mar	Dropped	Deferred to make room for add-in
Testing complete	4 Apr	8 Apr	4-day delay from add-in scope
Evaluation complete	11 Apr	14 Apr	Delayed by 3 days

Table 11: Planned vs Actual Project Milestones

Three things changed. The Python-to-TypeScript migration happened in week 3 after silent type mismatches across the API boundary caused runtime failures. The migration took four days but removed a whole category of bugs. The PowerPoint add-in was not in the original scope. Usability testing in week 7 revealed that the web-only download workflow created enough friction to offset the productivity gains. The add-in took three weeks to build but changed the system from a standalone generator into an assistant inside PowerPoint. Dropping RAG search (FR15, a ‘Could have’) made room for it. The iterative approach made these changes possible: each cycle had a working prototype, so scope changes could be assessed against a running system (SE7, SE8).

2.8 Risk Assessment

Table 12 identifies ten risks across security, reliability, and project delivery.

ID	Risk	Description
R1	Pitch Book Data Breach	A flaw in Supabase row-level security policies could expose one user's pitch books to another. In investment banking, pitch books often contain deal terms, valuations, and financial projections that are highly confidential.
R2	Financial Figure Hallucination	GPT-4o could insert fabricated market caps, revenue figures, or growth rates into slides. An analyst presenting incorrect numbers to a client could damage the firm's credibility and the deal itself.
R3	Malicious Template Upload	The template analyser parses uploaded .pptx files (ZIP archives containing XML) using JSZip and xml2js. A crafted file could attempt zip bomb decompression, path traversal, or oversized XML payloads to crash the server.
R4	Yahoo Finance Endpoint Change	yahoo-finance2 scrapes Yahoo's unofficial endpoints rather than calling a documented API. Yahoo has changed these endpoints before without notice, which would break financial data retrieval mid-generation.
R5	OpenAI Credential Exposure	The Express backend stores the OpenAI API key as an environment variable on Render. If the deployment is compromised or the key appears in server logs, an attacker could run up API costs under the project's account.
R6	Content Plan Schema Failure	GPT-4o occasionally returns malformed JSON or nests the content plan inside an unexpected wrapper object. If the Zod validation and unwrapping logic both fail, the entire generation pipeline stops.
R7	Slide Preview Exposure	LibreOffice converts generated PPTX files to PNG preview images stored in Supabase Storage. If the storage bucket defaults to public access, anyone with a direct URL could view confidential slide content without logging in.
R8	Platform Dependency Outage	The generation pipeline chains Vercel (web), Render (backend), Supabase (database and storage), and OpenAI (LLM). A failure at any point blocks generation, and there is no self-hosted fallback.
R9	Scope Creep Beyond Timeline	The project has a fixed 13-week timeline. Adding unplanned features risks delaying or cutting core deliverables. This materialised when the PowerPoint add-in was added and RAG search was dropped.
R10	Add-in Cross-Origin Session Breakage	The PowerPoint add-in runs inside an Office.js iframe on a different origin from the backend. Authentication depends on SameSite=None; Secure cookies, which behave differently across browser engines and Office versions. A platform update could silently break login.

Table 12: Risk assessment for the pitch deck generation system.

2.9 Severity Evaluation

Each risk was scored using Binary Risk Analysis (BRA) (Sapiro, 2011). BRA uses ten yes/no questions per risk (six for likelihood, four for impact) fed through lookup matrices to produce a deterministic Low/Medium/High classification. Table 13 summarises the results. Yahoo Finance endpoint changes (R4) and platform outages (R8) scored High because both involve external dependencies with no fallback. Six risks scored Medium. Two scored Low: malicious templates (R3) and credential exposure (R5), where mitigations reduce both likelihood and impact.

ID	Risk	Likelihood	Impact	Overall
R1	Pitch Book Data Breach	Low	High	Medium
R2	Financial Figure Hallucination	Medium	Medium	Medium
R3	Malicious Template Upload	Low	Low	Low
R4	Yahoo Finance Endpoint Change	High	Medium	High
R5	OpenAI Credential Exposure	Low	Low	Low
R6	Content Plan Schema Failure	Medium	Medium	Medium
R7	Slide Preview Exposure	Low	High	Medium
R8	Platform Dependency Outage	High	High	High
R9	Scope Creep Beyond Timeline	Medium	Medium	Medium
R10	Add-in Session Breakage	Medium	Medium	Medium

Table 13: BRA severity results. Full worksheet and matrix workings in Appendix 5.14.

2.10 Mitigation

R1 (data breach): Supabase row-level security restricts each user to their own records. R2 (hallucination): the data retrieval service passes real numbers directly to GPT-4o, and Zod validates every response. R3 (malicious templates): the backend validates ZIP structure, caps slide count, and disables external entity resolution. R4 (Yahoo Finance): the data retrieval service wraps yahoo-finance2 behind an abstract interface, so switching providers requires changing one file. R5 (credentials): keys sit in Render’s environment variable manager, never in source code. R6 (schema failure): an unwrapping step strips known JSON envelope patterns before Zod validation. R7 (preview exposure): signed URLs with expiry times; the storage bucket is private. R8 (platform outage): the architecture degrades gracefully on partial failures, but the dependency on four external services is permanent. R9 (scope creep): materialised when the add-in replaced RAG search. R10 (session breakage): fixed by setting SameSite=None; Secure cookies and configuring CORS for the add-in’s origin.

2.11 Ethics and Compliance

AI in financial services raises concerns around transparency, accountability, and human control (Jobin et al., 2019). The system addresses this through HITL design: generated content is marked as AI-assisted, and professional review is mandatory before distribution. Table 14 summarises data governance. These practices follow GDPR’s data minimisation principle (Bank for International Settlements, 2024).

Data Type	Storage	Retention	Deletion Trigger
User email and password hash	Supabase Auth	Indefinite	Account deletion
Uploaded .pptx templates	Supabase Storage	Indefinite	User deletes template
Template structural schema	Supabase DB (JSON)	Indefinite	Parent template deleted
Financial data (Yahoo/EDGAR)	In-memory cache	24 hours	Cache expiry
Generated .pptx files	Supabase Storage	Indefinite	User deletes pitch book
Slide preview PNGs	Supabase Storage	Indefinite	Parent pitch book deleted
Chat messages	Supabase DB	Indefinite	User deletes pitch book
LLM prompts and responses	Server logs	Per evaluation run	Tied to Git commit

Table 14: Data Governance: Storage, Retention, and Deletion Policies

OpenAI permits academic use of generated outputs. Yahoo Finance data comes through an unofficial package that would need replacing with a licensed provider for production. SEC EDGAR is public with no licensing restrictions. The HITL architecture meets EU AI Act requirements for human oversight (Bank for International Settlements, 2024).

2.12 Project Tracking

Progress is tracked using a Kanban board (SE8, SE10). Figure 3 shows the board near the end of the project.

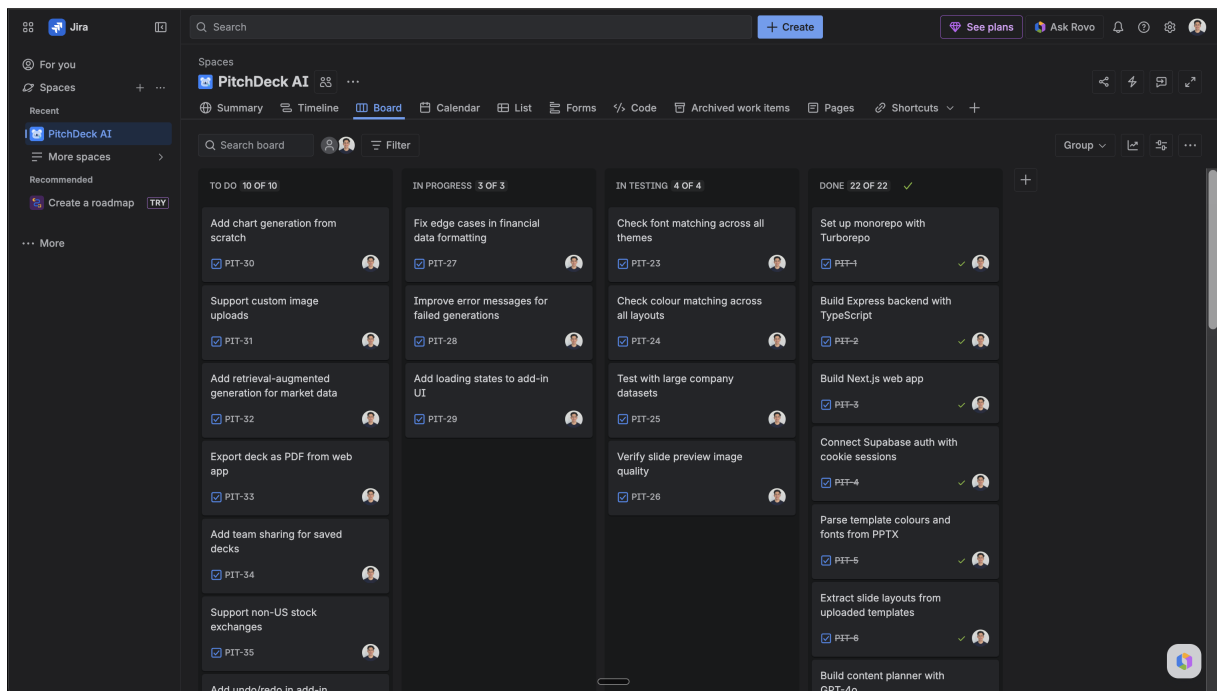


Figure 3: Kanban board showing task progress across four stages: To Do, In Progress, Testing, and Done.

3 Preliminary Research and Design Documentation

3.1 Literature Review

The project scope covers pitch book generation: taking a template, pulling financial data, planning content with an LLM, and assembling slides. RAG search over precedent materials was dropped to make room for the PowerPoint add-in (Section 2). The literature review focuses on what the system actually does, covering LLM productivity evidence, automated presentation generation, agentic architectures, and structured output reliability. Sources come from Google Scholar, ACM Digital Library, and IEEE Xplore, with snowball searching from Zheng et al. (2025) and L. Wang et al. (2024).

3.1.1 LLM Productivity

Large-scale studies back up LLM productivity gains with real numbers. Brynjolfsson et al. (2023) studied 5,179 customer support agents and found a 14% average productivity increase, with novice workers gaining up to 34%. Noy and W. Zhang (2023) found professionals completed writing tasks 40% faster with 18% higher quality when using ChatGPT. Both studies show uneven benefits: less experienced workers gain most because AI gives them access to patterns that experts have already learned.

Dell’Acqua et al. (2023) ran a randomised experiment with 758 BCG consultants and found that those using GPT-4 completed 12.2% more tasks, 25.1% faster, with 40% higher quality. Their main finding was the ‘jagged technological frontier’: tasks inside the AI’s capability boundary saw large gains, while tasks outside it made consultants perform worse. PB formatting and data retrieval sit inside the frontier. Analytical judgement sits outside it, which is why the system produces drafts rather than finished documents. Peng et al. (2023) found that developers with GitHub Copilot completed tasks 55.8% faster, with less experienced developers benefiting most. The parallel is direct: junior analysts are the less experienced workers who stand to gain the most.

That said, none of these studies addressed template-constrained document generation where outputs must match precise visual specifications. General writing assistance helps with content but does nothing for the mechanical formatting work that consumes analyst time. This project targets that gap.

3.1.2 Automated Presentation Generation

Table 15 breaks down the main approaches to presentation generation, measured against IB requirements. Every existing approach fails on at least one dimension that IB documents require.

Approach	Mechanism	Strengths	Limitations for IB
Rule-based layout	Algorithmic text-to-slide conversion	Fast, deterministic	Poor visual quality, no domain awareness
Rigid templates	Fixed layouts with manual content	High visual fidelity	Cannot adapt to varying content lengths
LLM-based (PPTAgent)	Reference analysis + edit-based generation	Preserves source style, good content	Domain-generic, no financial data
Multi-agent semantic (Di Genio)	Template-constrained multi-agent pipeline	Maintains document structure	Text documents only, not PowerPoint
VLM-based	Visual understanding of layouts	Handles complex visual elements	Approximate values, not exact matching

Table 15: Presentation Generation Approaches Compared

Hu and Wan (2013) introduced PPSGen, one of the first systems to learn slide generation from paper-slide pairs, but it produced rigid, text-heavy output. Yang et al. (2025) proposed Auto-Slides, a multi-agent system closer to this project’s pipeline approach, but targeting academic content.

PPTAgent (Zheng et al., 2025) comes closest. Its two-stage approach (analyse reference, generate matching slides) preserves the source deck’s visual identity. PPTAgent outperforms earlier systems on both content and design quality. The limitation is that it targets general-purpose presentations with no financial data integration or IB narrative conventions. Liu et al. (2024) surveyed 51 professionals and found template consistency was their top priority, which backs the template-first direction. None of these tools add domain awareness.

Vision-language models offer an alternative path. Faysse et al. (2024) show VLMs can understand document layouts visually. But code-based extraction via xml2js provides exact colour hex values and font sizes, while VLM-based approaches yield approximations. For corporate template matching where a colour being off by a shade flags the document as non-compliant, exact values are non-negotiable. This is why the project uses code-based extraction over visual understanding, building on PPTAgent’s template-first approach while adding the IB domain layer that PPTAgent lacks.

3.1.3 Agentic Architectures and Tool Use

L. Wang et al. (2024) propose a framework where agents operate through planning, memory, and action loops. This maps to PB generation: the system plans slide structure, remembers template constraints, and acts through tool calls and slide assembly.

Guo et al. (2024) show multi-agent systems add value when sub-tasks require negotiation or parallel reasoning. PB generation is sequential (analyse template, plan content, assemble slides), so a single agent with tool-use capabilities is sufficient. Qu et al. (2025) identify function calling as the primary LLM-tool interface, and Li et al. (2025) confirm it as the most broadly applicable pattern. In this project, the agent calls PptxGenJS to write slides and uses structured output schemas to maintain format consistency.

3.1.4 Structured Output Reliability

Getting reliable structured output from LLMs is a harder problem than it first appears. Shorten et al. (2024) benchmarked LLMs' ability to produce valid JSON and found an average success rate of 82.55%, varying from 0% to 100% by task. Geng et al. (2025) showed that constrained decoding helps but success rates drop with schema complexity. The content plan schema in this project is moderately complex: nested slide specifications with varying content block types. These findings shaped the validation strategy: every response is validated against a Zod schema, with a fallback generic plan on failure.

3.1.5 Hallucination and Human Oversight

Huang et al. (2024) distinguish two types of hallucination. Intrinsic hallucination contradicts information given in the prompt. Extrinsic hallucination generates claims with no grounding in any source. For IB, extrinsic hallucination is more dangerous: made-up numbers or invented deal precedents could reach clients and regulators, causing regulatory problems or legal liability.

Alansari and Luqman (2025) group prevention into three categories: grounding outputs in real sources, forcing structured output formats, and human review. No single strategy eliminates hallucination. This project layers all three: financial data comes from APIs rather than LLM generation, structured JSON schemas enforce the format of each slide specification, and human review is mandatory before any output reaches a client.

Wu et al. (2022) classify HITL approaches into data processing, interventional training, and system design. This project uses system design: the analyst defines inputs, reviews outputs, and decides whether to use them. The EU AI Act requires human oversight for AI in financial services (Bank for International Settlements, 2024), making this a compliance requirement as well as a design choice. Junior analysts already produce drafts that senior bankers revise. The system replaces the blank page, not the review process.

3.1.6 Research Gaps

Four gaps emerge from this review. First, no presentation generation tool targets IB-specific conventions. PPTAgent (Zheng et al., 2025), Auto-Slides (Yang et al., 2025), and PPSGen (Hu and Wan, 2013) all target general or academic presentations. Second, no system automates both structure and content within a single pipeline (Achachlouei et al., 2023). Third, no system integrates data retrieval with template-aware assembly. Fourth, the productivity literature measures general tasks but none measured template-constrained document generation where outputs must match precise visual specifications.

This project targets these gaps with code-based template extraction for exact visual matching, IB-specific content planning, and mandatory human review for accuracy. The evaluation measures template matching and generation speed, and includes an informal usability evaluation with junior analysts.

3.2 System Design

This section documents the system architecture, data models, and component specifications derived from the requirements and literature review. The design puts template matching, clean separation of components, and human oversight first.

3.2.1 Architecture Overview

Figure 4 shows the high-level system architecture. The design follows a pipeline pattern with three main stages: template analysis, content planning, and slide assembly.

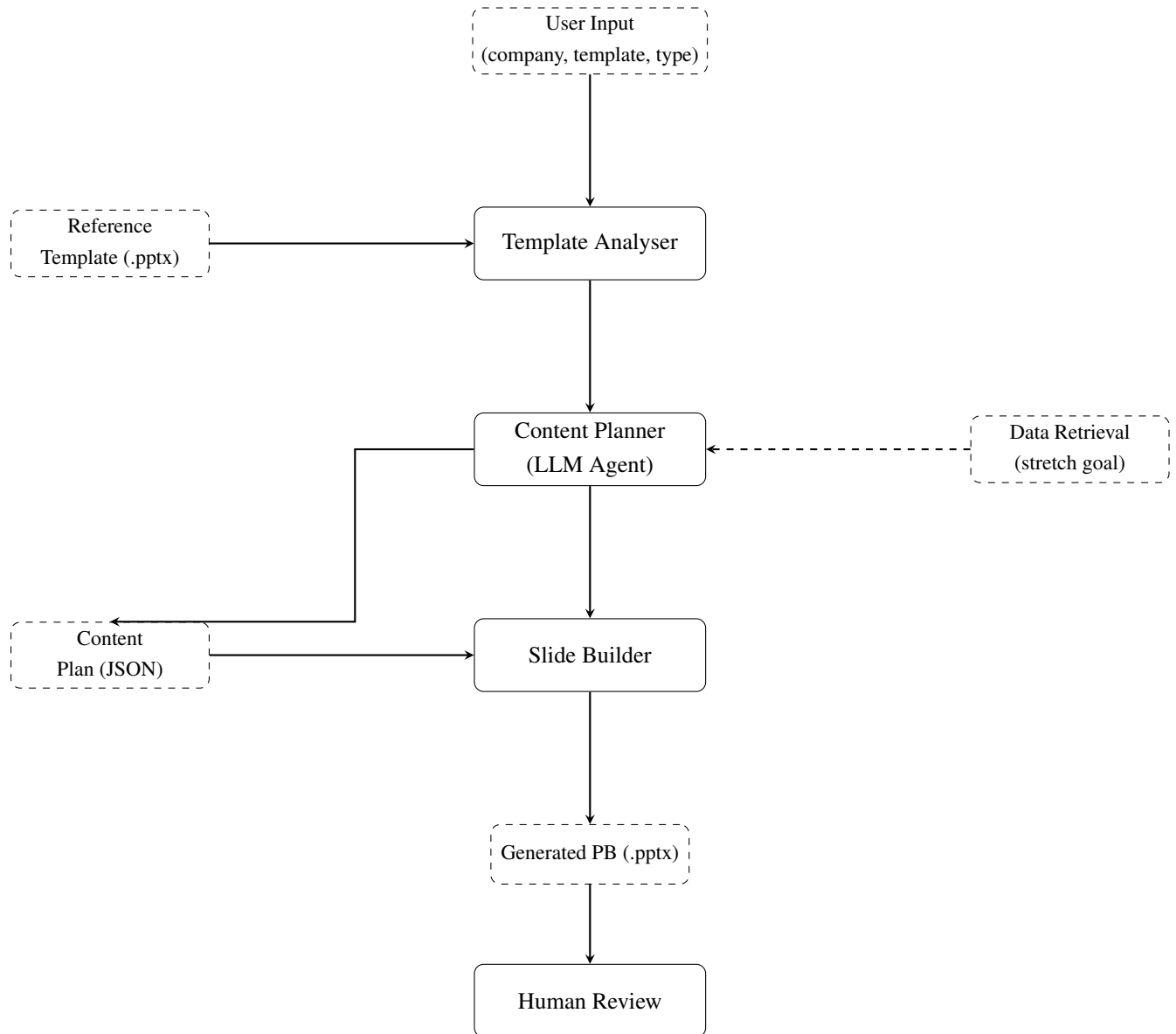


Figure 4: System Architecture showing the three-stage core pipeline. Data Retrieval is a stretch goal.

The architecture separates concerns into independent modules (SE2, SE9). The Template Analyser extracts layout patterns without generating content. The Content Planner uses an LLM to structure the narrative without touching PowerPoint files. The Slide Builder assembles files from specifications without making content decisions. This separation supports testing and allows components to be reused independently. Data Retrieval from external APIs is a stretch goal that can be added later without changing the core pipeline.

3.2.2 Data Models

The system uses two primary data structures that flow between components. Table 16 lists the fields.

Model	Key Fields	Purpose
TemplateSchema	layouts[], colours, fonts, placeholders[]	Stores extracted template properties for content planning and slide assembly
SlideLayout	layout_index, placeholder_map, background	Maps placeholder positions to content types for a single layout
ContentPlan	slides[], metadata	LLM-generated specification of slide content and structure
SlideSpec	layout_id, placeholders, charts[], tables[]	Content specification for a single slide

Table 16: Primary Data Model Specifications

Template Schema captures extracted template properties including layouts, colour palettes, fonts, and placeholder specifications. Each slide layout maps placeholder indices to content types (title, body, picture, chart, table).

Content Plan specifies the content for each slide: which layout to use, what text goes in each placeholder, and what data to display. The LLM generates this structure. The Slide Builder consumes it.

Figure 5 shows the database schema. Row-level security (RLS) policies on every table restrict access so that authenticated users can only read and write their own records.

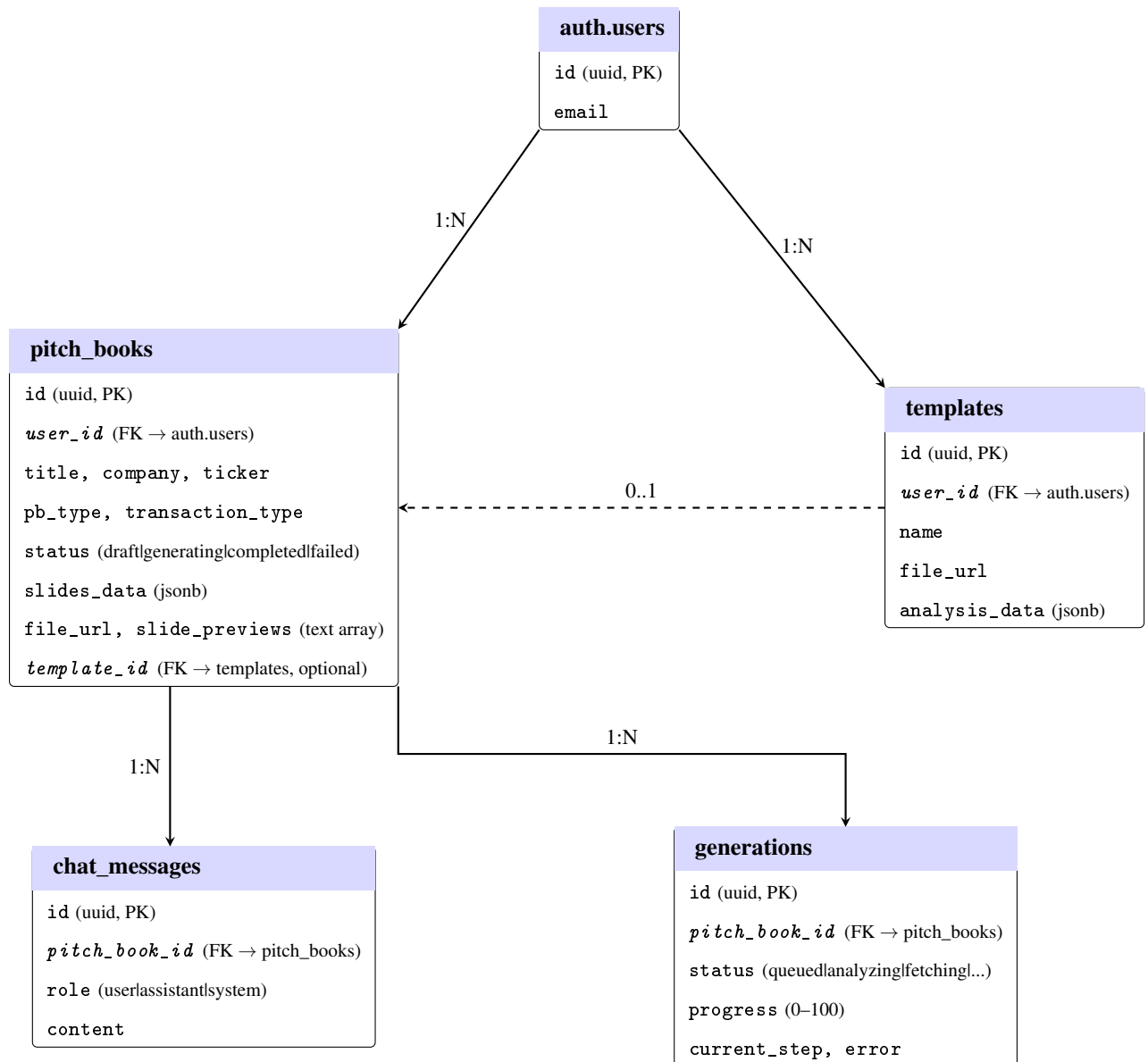


Figure 5: Database entity-relationship diagram. Primary keys in bold, foreign keys in italics. All tables use UUID primary keys with row-level security. The dashed line indicates an optional foreign key (pitch books may or may not reference a template).

3.2.3 Component Specifications

Template Analyser. Input: reference .pptx file. Output: TemplateSchema JSON. Uses JSZip and xml2js to parse the .pptx file at the XML level, extracting slide masters, layouts, colour schemes, and placeholder positions. Identifies placeholder types (title, body, picture, chart, table) by inspecting shape properties. Extracts exact colour values as hex codes and font specifications including family, size, and weight.

Content Planner. Input: user parameters, TemplateSchema. Output: ContentPlan JSON. Implemented as an LLM call with JSON schema validation to check the output structure. Selects appropriate layouts from the template schema based on content requirements. Plans slide sequence following IB narrative conventions (context, analysis, recommendation).

Slide Builder. Input: ContentPlan, TemplateSchema, reference template. Output: .pptx file. Uses PptxGenJS to generate slides with code. Iterates through ContentPlan slides, selects layouts, and populates placeholders. Applies exact formatting from TemplateSchema. Generates charts and tables with code.

3.2.4 API Design

The backend exposes a REST API via Express.js. The primary endpoint accepts generation requests and returns job identifiers for async processing.

Listing 1: Pitch book creation request interface (from shared-types)

```

1 // POST /api/pitchbooks - returns pitch book record with generation
  queued
2 interface CreatePitchBookRequest {
3   title: string;
4   company: string;
5   ticker?: string;
6   transaction_type: 'ma' | 'capital_raising' | 'restructuring'
7                     | 'ipo' | 'debt_financing';
8   pb_type: 'company_overview' | 'market_update'
9           | 'transaction_summary';
10  template_id?: string; // reference template UUID
11  additional_context?: string; // free-text instructions
12  color_theme?: string; // e.g. 'navy_gold', 'teal_coral'
13  design_style?: string;
14 }

```

The full API is documented with an OpenAPI 3.0 specification and served through Swagger UI (Figure 6). Authenticated endpoints (marked with a lock icon) require a valid Supabase session cookie.

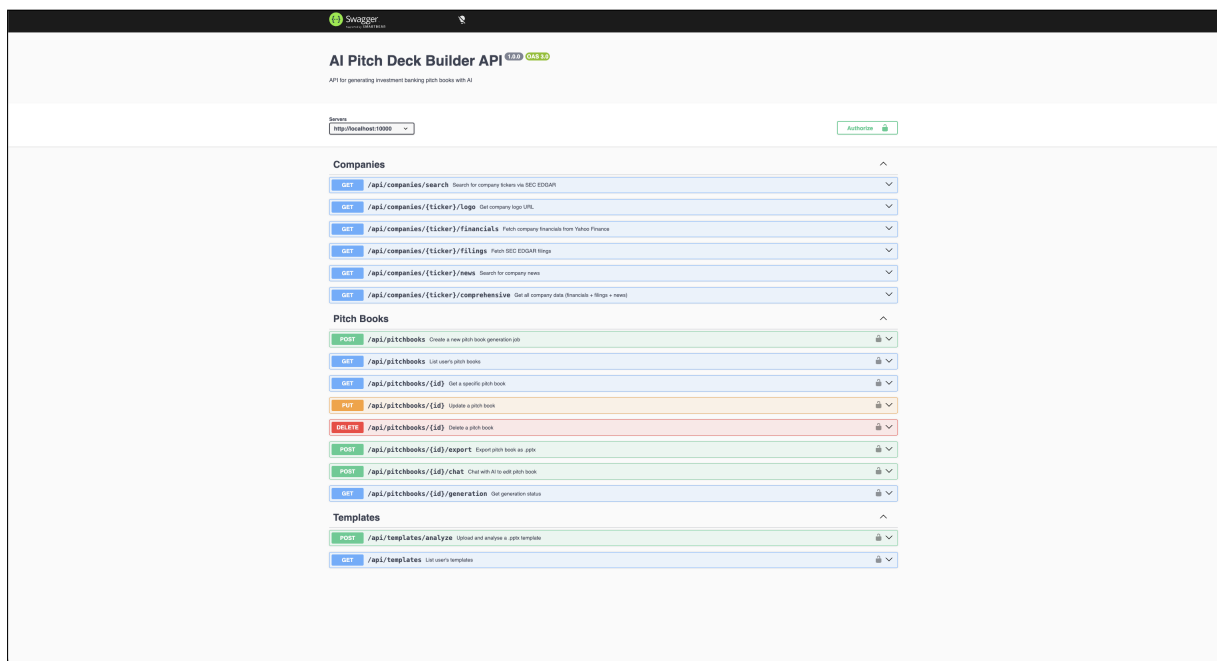


Figure 6: Swagger UI showing the REST API endpoints grouped by resource: Companies (data retrieval), Pitch Books (generation and CRUD), and Templates (upload and analysis).

3.2.5 Design Trade-offs

Table 17 summarises the four main design trade-offs.

Decision	Chosen Option	Rationale
Single vs multi-agent	Single agent with tools	PB generation is sequential; multi-agent overhead not justified
Programmatic vs VLM template extraction	Programmatic extraction (JSZip + xml2js)	Exact values needed for corporate template matching
SDK vs custom orchestration	Structured LLM calls with JSON schemas	Vendor-agnostic; component interfaces allow swapping later
Sync vs async generation	Async job processing	Generation takes minutes; users poll for completion

Table 17: Design Trade-offs

3.2.6 Data Flow

Figure 7 shows how data moves through the system during a generation request.

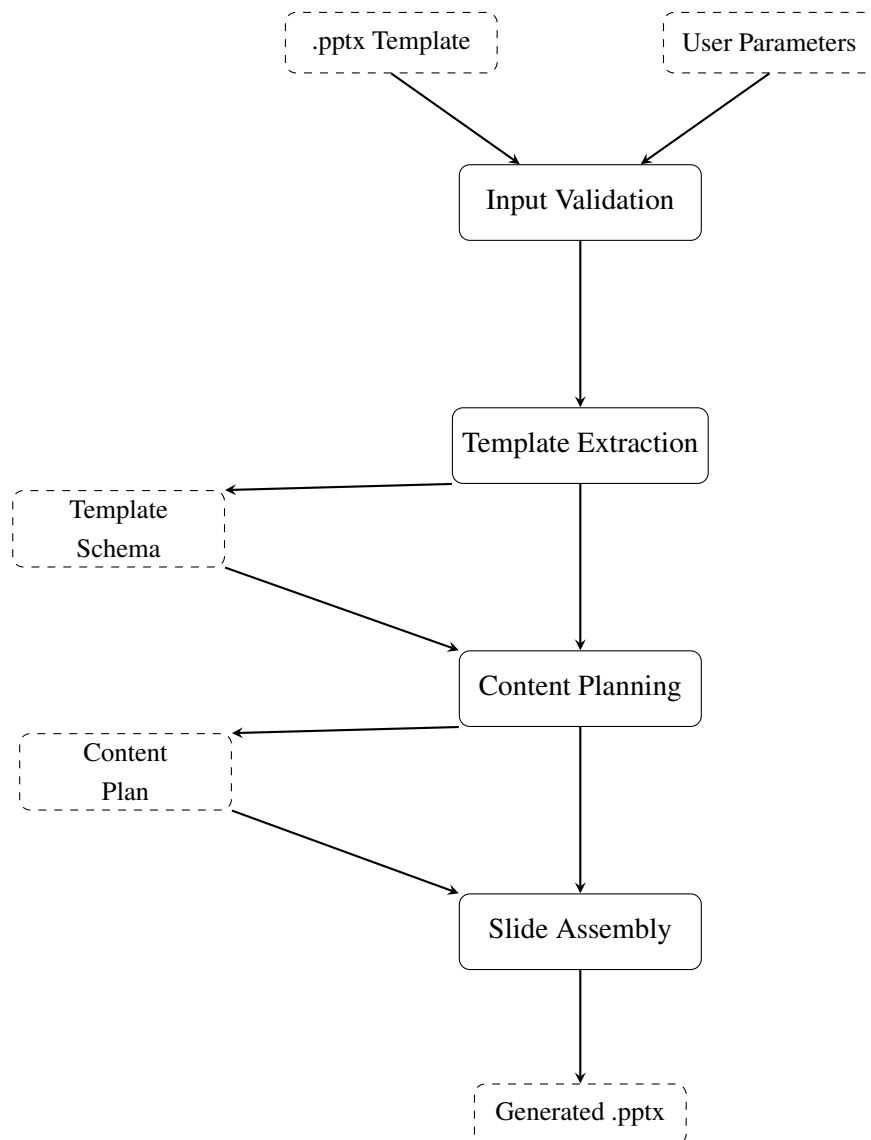


Figure 7: Data flow through the generation pipeline

3.2.7 Design Limitations

The system cannot generate charts from scratch. It can only populate existing chart placeholders. The template analyser handles standard placeholder-based layouts but does not reverse-engineer arbitrary PowerPoint constructions with overlapping elements or custom animations. The content planner has no memory across sessions. The system produces drafts, not finished documents: human review is mandatory (K6, SE1).

4 Project Implementation and Outcomes

This section covers what was built, how it works, and where things changed from the original plan.

4.1 Architecture Evolution

The original design called for a Python backend with FastAPI and a Next.js frontend. Two weeks in, this changed: every data structure crossing the API boundary had to be defined twice, and mismatches caused silent runtime failures. Switching everything to TypeScript (SE3) took four days but removed a whole category of bugs. The `shared-types` package now defines every interface once, and the compiler catches mismatches at build time.

4.2 Monorepo Structure

The codebase uses Turborepo to manage three applications and one shared package (SE12):

Package	Technology	Purpose
<code>apps/service</code>	Express.js, TypeScript	Backend API: orchestration, template analysis, content planning, slide building
<code>apps/web</code>	Next.js, React, TypeScript	Web frontend: PB management, slide preview, generation trigger
<code>apps/powerpoint-addin</code>	Vite, React, Office.js	PowerPoint taskpane: in-editor generation and AI-assisted editing
<code>packages/shared-types</code>	TypeScript	Shared interfaces, enums, and type definitions across all apps

Table 18: Monorepo Package Structure

Figure 8 shows the web app dashboard after login.

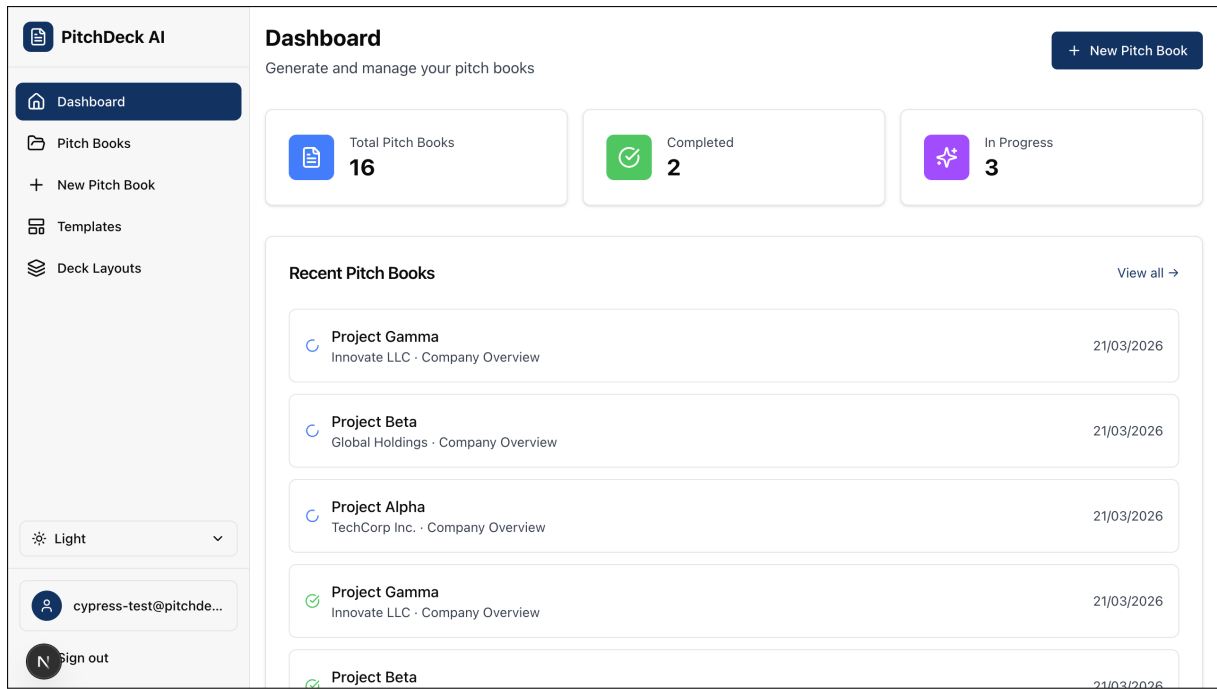


Figure 8: Web app dashboard showing pitch book counts, recent generation history, and sidebar navigation.

4.3 Orchestration Layer

The orchestrator runs each stage in sequence and stays thin: each service module handles its own domain. Figure 9 shows the full pipeline. If data retrieval fails, it passes an empty object to the content planner rather than aborting. The orchestrator passes the full template schema to GPT-4o rather than a summary, because early versions with only layout names produced worse output.

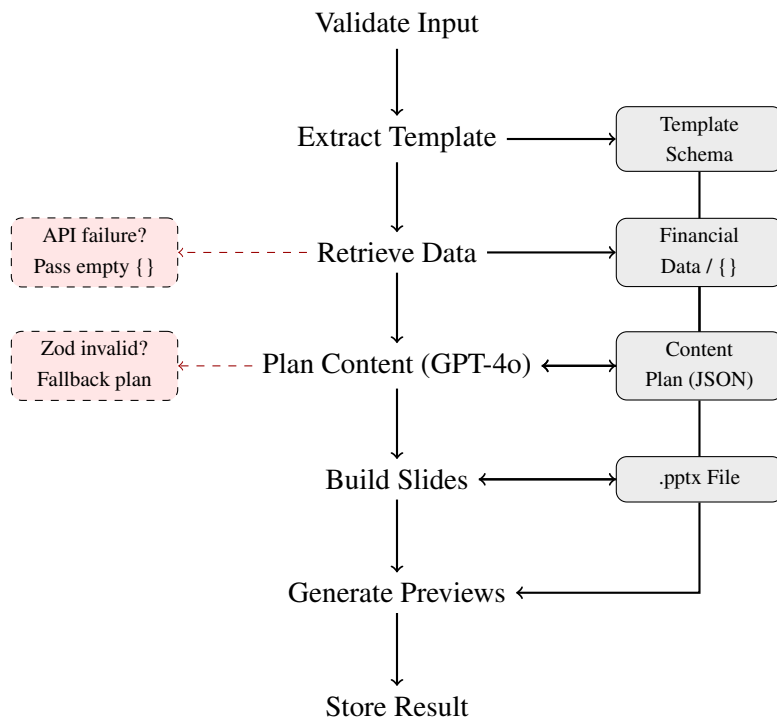


Figure 9: Orchestration pipeline. Each stage passes its output to the next. Dashed paths show graceful failure handling: data retrieval passes an empty object on API failure, and the content planner falls back to a generic plan if Zod validation fails.

4.4 Deck Layouts and Colour Themes

The system ships with seven deck layouts (Company Overview, Investor Pitch, Market Update, Transaction Summary, Industry Overview, Fundraising Deck, Due Diligence) and eight built-in colour themes. The layouts are hardcoded because IB slide sequences are standard across the industry. The visual style is customisable: users pick a theme or upload a bank's own template file. When a template is uploaded, the analyser extracts that bank's exact colours, fonts, and layout positions, and the slide builder applies them to whichever layout the user picks. Figure 10 shows the creation form.

The screenshot shows the 'New Pitch Book' form in the PitchDeck AI application. The left sidebar contains navigation links: Dashboard, Pitch Books, New Pitch Book (highlighted), Templates, and Deck Layouts. Below the sidebar, there is a theme selector set to 'Light', a user profile for 'cypress-test@pitchde...', and a 'Sign out' button. The main form area is titled 'New Pitch Book' with the instruction 'Fill in the details and let AI generate your pitch book.' It is divided into two sections: 'Basic Information' and 'Presentation Details'. The 'Basic Information' section includes a 'Pitch Book Title *' field with the value 'Project Delta Generation', a 'Company Name *' field with the value 'Stark Industries', and a 'Ticker Symbol' field with a search icon and the placeholder text 'Search ticker or company name...'. The 'Presentation Details' section includes a 'Pitch Book Type *' dropdown menu with 'Company Overview' selected, a 'Transaction Type' dropdown menu with 'Mergers & Acquisitions' selected, a 'Design Style' dropdown menu with 'Modern' selected, and a 'Color Theme' dropdown menu with 'Navy & Gold' selected. Below the 'Pitch Book Type' dropdown, there is a description: 'Business overview, financials, and market position'.

Figure 10: New pitch book creation form. The user selects a deck layout, colour theme, and enters company details before generating.

4.5 Core Pipeline

The generation pipeline follows the design from Section 3, implemented as a chain of service modules within the Express backend. Each stage is a separate TypeScript module with a defined input/output contract. Development followed a pattern: build the simplest version, test it against real templates, identify the failure mode, and fix it.

4.5.1 Template Analysis

The template analyser went through three iterations, each driven by a specific failure in the previous version (SE4).

Iteration 1: PptxGenJS only. The first attempt used PptxGenJS for both reading and writing slides. PptxGenJS can create slides from scratch but has no API for reading existing .pptx files. The only option was to generate new slides and hope the styling matched. It did not. Colours were wrong, placeholder positions were guessed rather than measured, and the output looked nothing like the input template. This approach was scrapped after two days.

Iteration 2: JSZip + xml2js (raw XML parsing). PowerPoint files are ZIP archives containing Office Open XML documents. JSZip opens the archive, and xml2js parses the XML inside. This gave direct access to every property: colours from ppt/theme/theme1.xml, layouts from ppt/slideLayouts/, and placeholder positions from shape definitions. The approach worked but came with a problem. PowerPoint's OOXML specification stores measurements in English Metric Units (EMU), where one inch equals 914,400 EMU. Colours can be stored as hex values, theme references (accent1), or system colour names. Placeholder types use numeric codes, but custom placeholders have no type attribute at all. Each of these required manual parsing and conversion logic.

Iteration 3: heuristics and fallbacks for analysis. Testing against real corporate templates exposed gaps in the raw XML approach. Custom placeholders with no type attribute made up roughly 30% of all placeholders in the test set. The analyser added position-based heuristics: a placeholder in the top 15% of the slide with large font settings is classified as a title, even without a type code. A wide placeholder in the bottom half with small fonts is treated as body text. This brought classification accuracy to 93% across test templates. When parsing fails entirely (corrupted file, unusual XML structure), the analyser falls back to professional defaults: a navy and gold colour scheme with standard IB layout patterns.

Listing 2: Template extraction core logic (simplified)

```

1  async function analyseTemplate(buffer: Buffer): Promise<TemplateSchema>
    {
2      const zip = await JSZip.loadAsync(buffer);
3
4      // Parse theme for colour palette
5      const themeXml = await zip.file('ppt/theme/theme1.xml')?.async('string
        ');
6      const theme = await parseStringPromise(themeXml);
7      const colours = extractColourScheme(theme);
8
9      // Parse each slide layout
10     const layouts: SlideLayout[] = [];
11     for (const [path, file] of Object.entries(zip.files)) {
12         if (path.match(/ppt\/slideLayouts\/slideLayout\d+\.xml/)) {
13             const xml = await file.async('string');
14             const parsed = await parseStringPromise(xml);
15             layouts.push(extractLayout(parsed, colours));
16         }
17     }
18
19     return { colours, fonts: extractFonts(theme), layouts };
20 }

```

Each version failed in a specific, observable way, and the fix was targeted rather than speculative (SE4). The unit tests grew alongside: the third iteration added edge-case tests for missing themes, custom placeholders, and corrupted archives (SE5).

The analyser reads layout information from the slide master. Figure 11 shows the master view for a sample template. Each thumbnail on the left is a slide layout. The master defines where placeholders go, what fonts to use, and what the background looks like. The analyser reads these from the ppt/slideLayouts/ directory and records the placeholder types, positions, and styling.

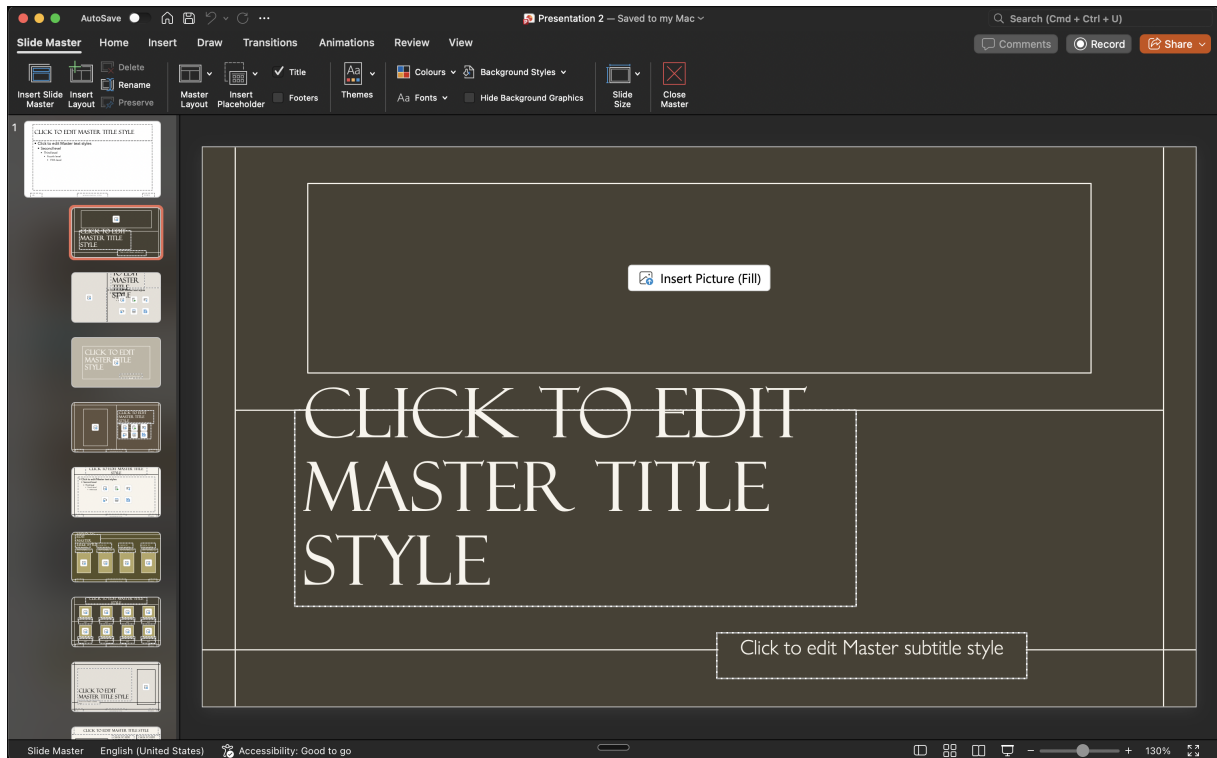


Figure 11: Slide master view showing the available layouts for a template. Each layout defines placeholder positions, fonts, and background styling that the generation pipeline inherits.

4.5.2 Content Planning with GPT-4o

The content planner sends user parameters and the extracted template schema to GPT-4o, requesting a structured content plan. The system prompt (Listing 3) bans generic phrasing and specifies minimum content density per slide type, because early versions produced boilerplate and sparse output.

Listing 3: Content planner system prompt excerpt (from pitchbook-content.prompt.ts)

```
1 export const SYSTEM_PROMPT = 'You are a senior Managing Director at
2 Goldman Sachs creating investment banking pitch books.
3
4 CRITICAL RULES:
5 1. ALL financial data MUST use the EXACT numbers provided in the
6   user message. NEVER fabricate numbers.
7 2. Every chart MUST have a "data" array with real numbers.
8 3. Content must be deeply specific to the company. Mention their
9   actual products, markets, competitors, recent events.
10 4. Write like a real IB analyst: precise, data-driven, no fluff.
11 5. NEVER write generic phrases like "leading company in its sector"
12    or "strong market position". Be SPECIFIC.
13
14 CONTENT DENSITY:
15 - Every Content Slide MUST have at least 5-7 bullet points.
16 - Every bullet point must be a FULL SENTENCE with specific data.
17   BAD: "Strong revenue growth".
18   GOOD: "Revenue grew 61% YoY to 130.5B in FY2024, driven by
19   Data Center segment expansion to 115.2B."
20 - Financial Tables should have 4-5 years of data with 5+ rows.
21
22 You MUST return JSON in EXACTLY this format:
23 { "title": "...", "narrative_arc": "...", "slides": [...] }';
```

Getting valid structured output was the main challenge (SE1). Shorten et al. (2024) found an average JSON success rate of 82.55% across LLM benchmarks. GPT-4o occasionally wraps responses in unexpected envelope objects. The content planner handles this with an unwrapping step before Zod (Colinhacks, 2024) schema validation. If validation fails, a hardcoded fallback plan keeps generation going. Across 12 evaluation runs, all produced valid JSON without needing the fallback. The prompt includes retrieved financial data so the LLM places real numbers rather than fabricating them.

Listing 4: Content plan validation with Zod (from pitchbook-content.schema.ts)

```

1  const contentBlockSchema = z.object({
2    type: z.enum(['heading', 'paragraph', 'bullet_list',
3                 'table', 'chart', 'metric']),
4    content: z.any(),
5  });
6
7  const slideSchema = z.object({
8    index: z.number().optional().default(0),
9    title: z.string(),
10   layout: z.string().optional().default('Content Slide'),
11   talking_points: z.array(z.string()).optional().default([]),
12   data_requirements: z.array(z.string()).optional().default([]),
13   content_blocks: z.array(contentBlockSchema).optional().default([]),
14 });
15
16 const contentPlanSchema = z.object({
17   title: z.string(),
18   narrative_arc: z.string().optional().default(''),
19   slides: z.array(slideSchema),
20 });
21
22 function parseContentPlan(raw: unknown): ContentPlanOutput {
23   return contentPlanSchema.parse(raw);
24 }

```

4.5.3 Slide Building

The system has two slide building paths, selected by whether the user uploaded a template (SE10).

Default mode uses PptxGenJS to generate slides from scratch with a built-in theme selected by pitch book type. Theme colour references (accent1, dk1) are resolved to hex values before applying, since PptxGenJS has no theme colour system.

Template mode clones the template's actual slides and replaces placeholder text using regex on the raw XML. Figure 12 shows a sample template with 14 slides. The builder parses each slide's .rels file to find its layout name, then for each content plan slide, picks the best-matching template slide. It finds each <p:sp> shape containing a <p:ph> (placeholder) element and replaces the <p:txBody> content while preserving the original <a:bodyPr>, <a:lstStyle>, and <a:rPr> (font, size, colour) via regex extraction. The result: generated text comes out in the template's own font, size, and colour.

This approach uses regex on the raw XML rather than xml2js round-tripping, which corrupted OOXML namespace declarations and produced files that Office.js rejected. The builder also injects <a:normAutofit> to auto-shrink overflowing text and truncates content per placeholder type (120 characters for titles, 7 bullets at 150 characters each, 500 for body text).

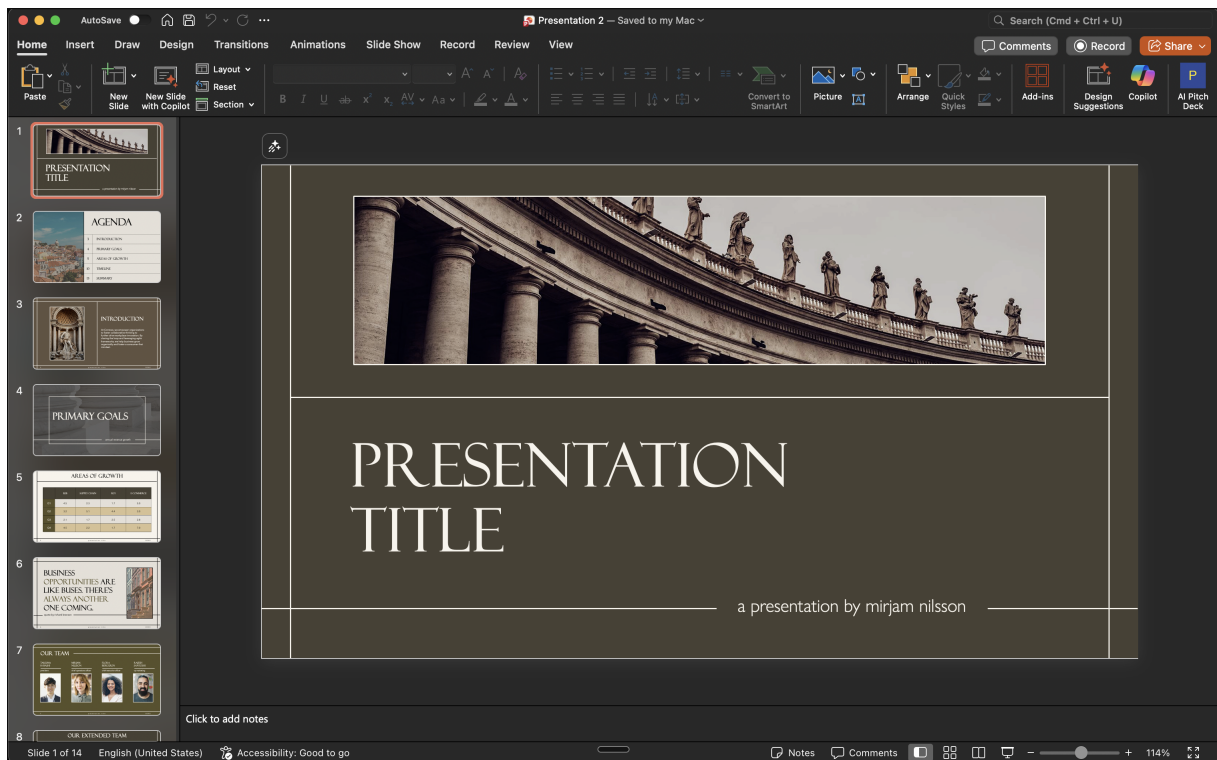


Figure 12: Original template with 14 slides across different layout types, each with a consistent brown and gold visual style.

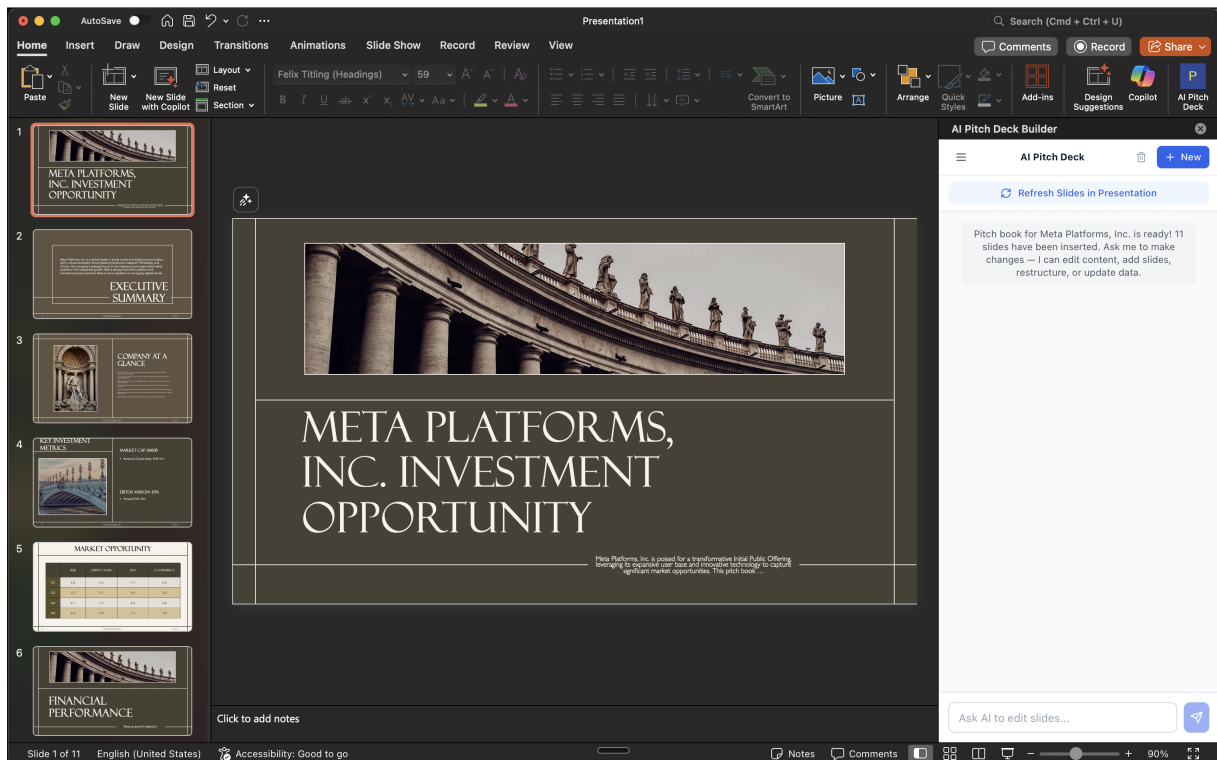


Figure 13: Generated output for Meta Platforms using the template from Figure 12. The 11 slides inherit the template's Felix Titling font, brown background, and gold accents. The add-in taskpane (right) shows slides were inserted directly into PowerPoint.

4.5.4 Data Retrieval

The data retrieval service pulls financials from Yahoo Finance (`yahoo-finance2`) and SEC EDGAR (K5), caching responses for 24 hours. If an API call fails, the service returns graceful null values so generation proceeds with reduced data rather than failing entirely.

4.5.5 Slide Preview Generation

The system uses LibreOffice in headless mode to convert .pptx to PDF, then `pdftoppm` to render each page as a PNG. Previews are uploaded to Supabase Storage and displayed in the web frontend. Neither this pipeline nor the add-in below were in the original design. Both emerged from usability problems found during implementation.

4.6 PowerPoint Add-in

A walkthrough in week 7 exposed the flaw in the web-only design: fixing one slide required regenerating the full deck and manually copying it back. The round trip took eight minutes for a one-sentence change. The solution was a taskpane add-in built with Office.js that runs inside PowerPoint, communicating with the same Express backend.

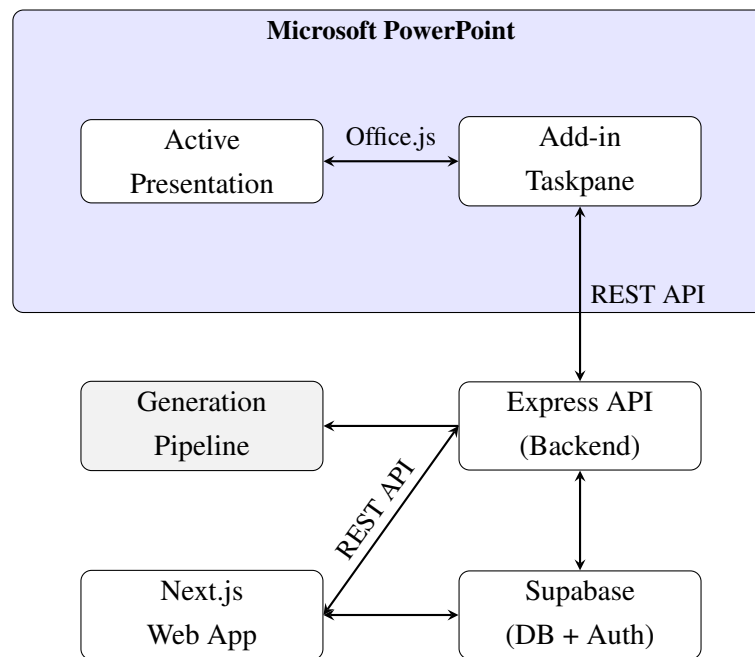


Figure 14: System architecture with PowerPoint Add-in. The add-in and web app share the same backend and database. Generated slides are pushed directly into the active presentation via Office.js.

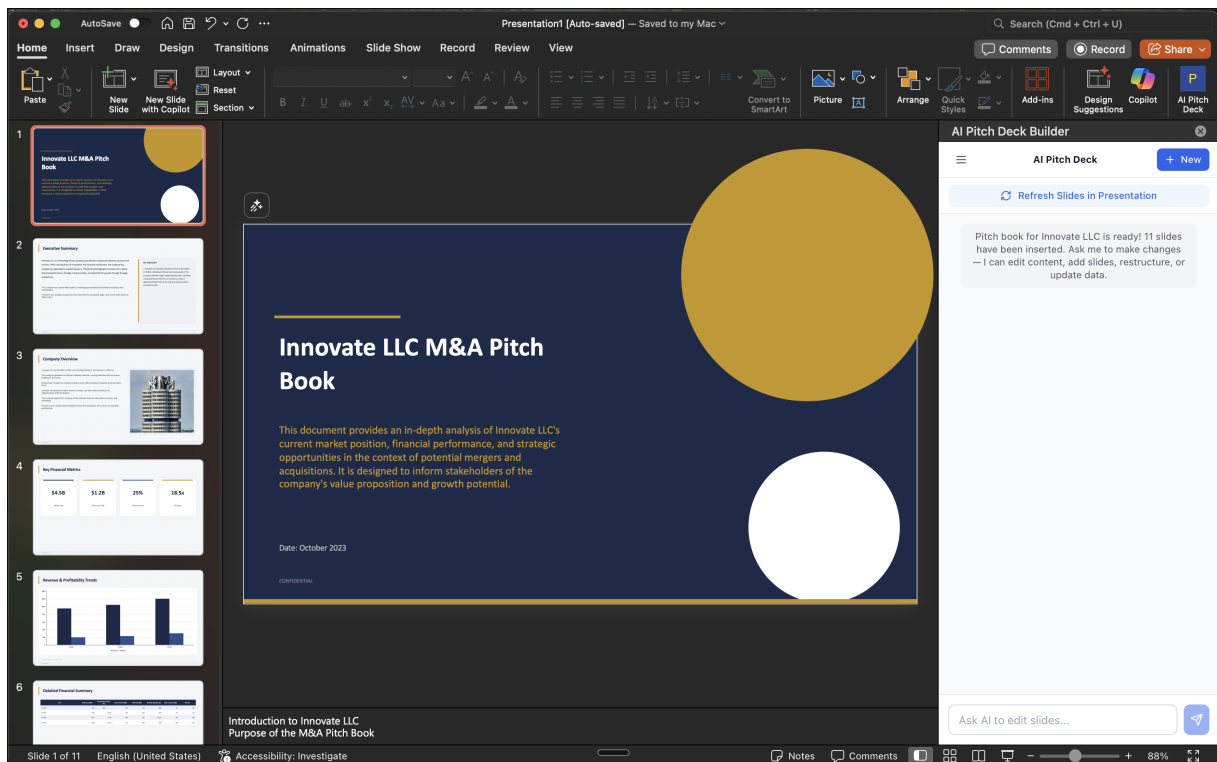


Figure 15: PowerPoint with the add-in taskpane open. Generated slides appear in the slide panel on the left, and the AI chat input sits at the bottom of the taskpane.

The `insertSlidesFromBase64()` API pushes the generated .pptx directly into the active presentation. The web app became a management interface for browsing pitch books. Both applications share the same backend (K5).

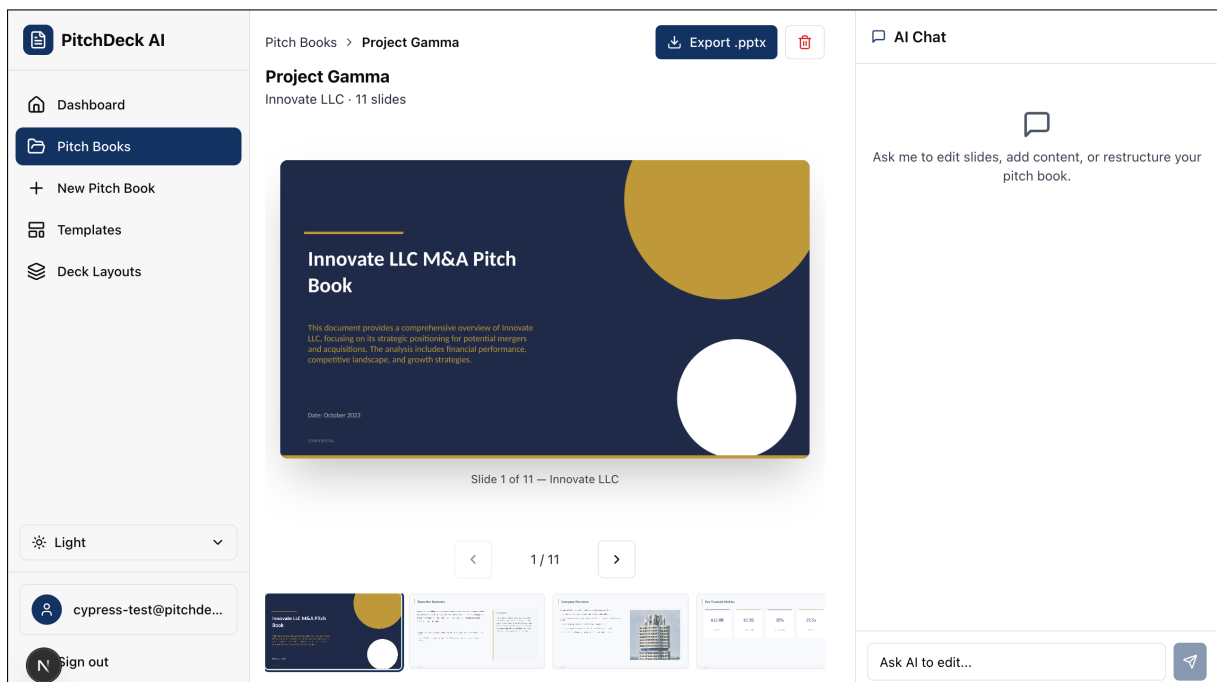


Figure 16: Pitch book viewer showing slide preview, thumbnail navigation, and AI chat panel.

4.7 AI Chat Assistant

The pitch book viewer includes an AI chat panel (visible in Figure 16) that lets users request slide changes in plain English. The chat runs on GPT-4o-mini with a system prompt that includes each slide’s title and layout type. Messages are stored in a `chat_messages` table, preserving history across sessions. The chat is advisory: the assistant provides instructions, but the user applies the edits.

4.8 Authentication and Session Management

Supabase Auth handles user authentication across both applications. The Express backend acts as the single auth source, checking session cookies on every request (SE1). The add-in runs inside an iframe, which blocks cross-origin cookies by default. This required `SameSite=None`; Secure cookie attributes and explicit CORS configuration.

4.9 Testing

The project uses three levels of testing: unit tests for individual modules, integration tests for service interactions, and end-to-end (E2E) tests for full user flows (SE5, SE11).

4.9.1 Unit and Integration Tests

Vitest (Vitest Team, 2024) runs all unit and integration tests. It was chosen over Jest for native Vite module resolution and TypeScript support without extra setup. Turborepo runs all test suites in parallel via `turbo run test` and skips unchanged packages on repeat runs. Table 19 shows what each test suite covers.

App	Test File	What it Tests
apps/service	template-analyser.test.ts	Colour, font, and layout extraction from known .pptx fixtures
apps/service	content-planner.test.ts	Zod validation rejects malformed LLM output, retries on failure
apps/service	slide-builder.test.ts	Output .pptx has correct formatting and valid XML structure
apps/service	default-themes.test.ts	Theme colours match specs when no template is uploaded
apps/service	auth-middleware.test.ts	Session validation, cookie parsing
apps/web	api.test.ts	API client calls and error handling
apps/web	use-unsaved-changes.test.ts	Hook warns before navigating away
apps/powerpoint-addin	office-helpers.test.ts	Office.js mock interactions
apps/powerpoint-addin	component-logic.test.ts	Form state and validation logic

Table 19: Unit and Integration Test Coverage

Tests use `vi.mock()` to stub external dependencies (SE11) so they run without API keys or a database connection. Figure 17 shows the Vitest UI after a full run with all 134 tests passing.

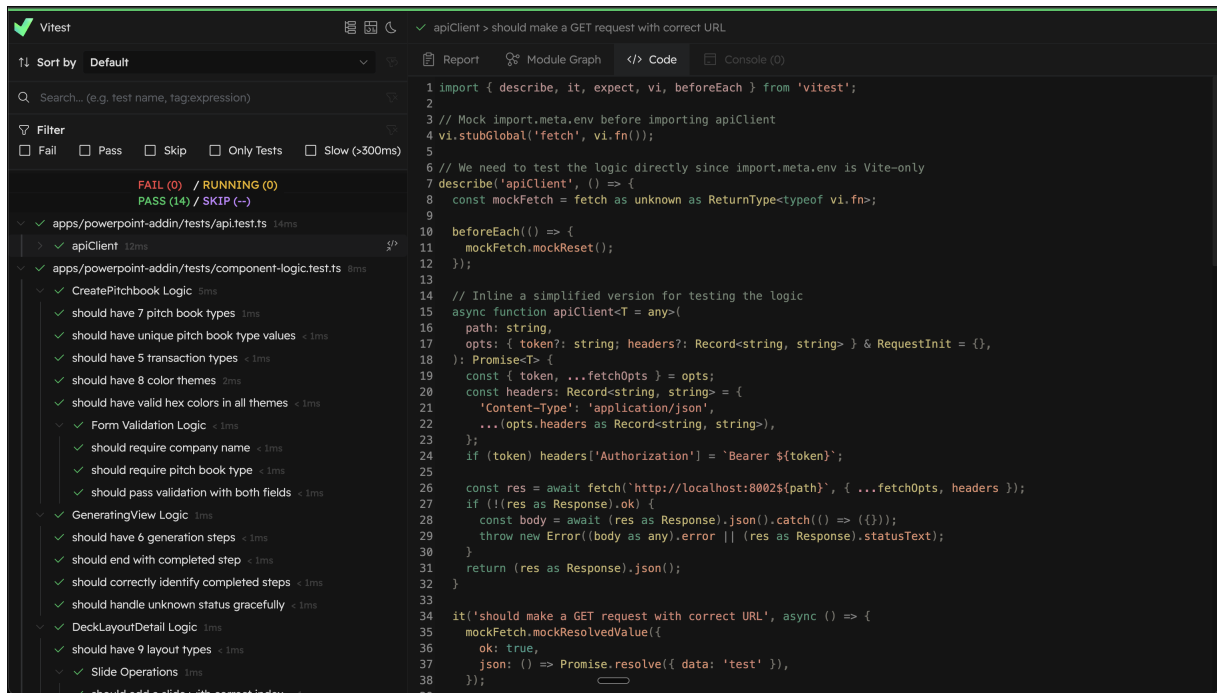


Figure 17: Vitest UI showing all 14 test files and 134 tests passing across the three apps. The right panel displays the selected test’s source code and can switch to console output on failure.

4.9.2 E2E Tests

Cypress (Cypress.io, 2024) was chosen over Playwright for its low setup time with Next.js and built-in time-travel debugger. Tests run against a real Supabase backend with a test user account. Seven spec files cover the main user flows:

Spec File	What it Tests
auth.cy.ts	Login, logout, session persistence
dashboard.cy.ts	Dashboard loads, shows recent PBs
pitchbooks.cy.ts	Create, view, delete pitch books
templates.cy.ts	Upload and manage templates
landing.cy.ts	Public landing page renders correctly
layouts.cy.ts	Responsive layout across screen sizes
settings.cy.ts	User settings and preferences

Table 20: Cypress E2E Test Specs

Figures 18 to 20 show the Cypress test runner in action.

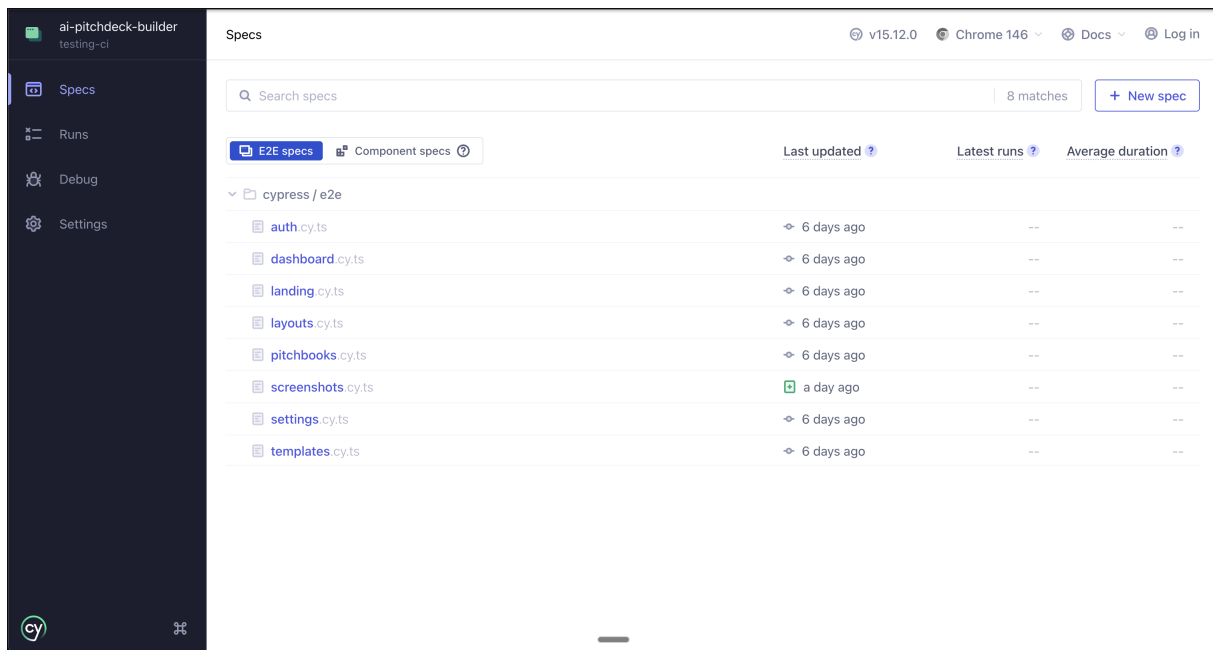


Figure 18: Cypress test runner showing the E2E spec files grouped by feature area.

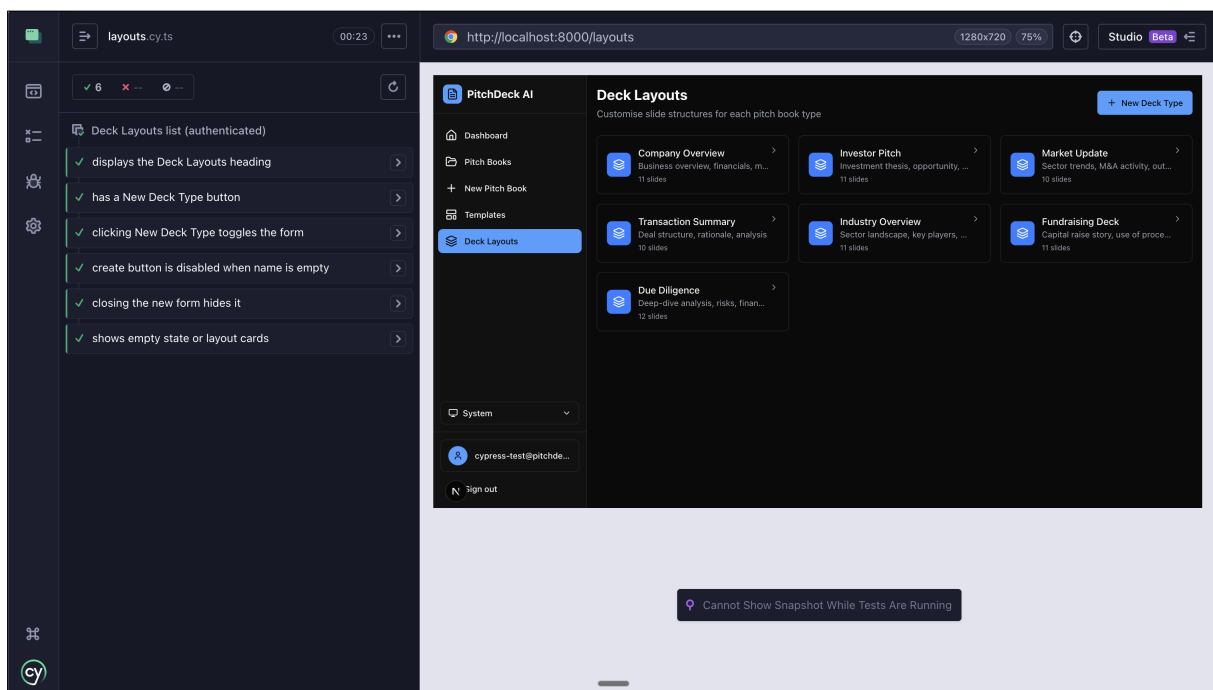


Figure 19: A passing Cypress test. The left panel shows each automated step (clicking elements, checking text), and the right panel shows the browser state at each step.

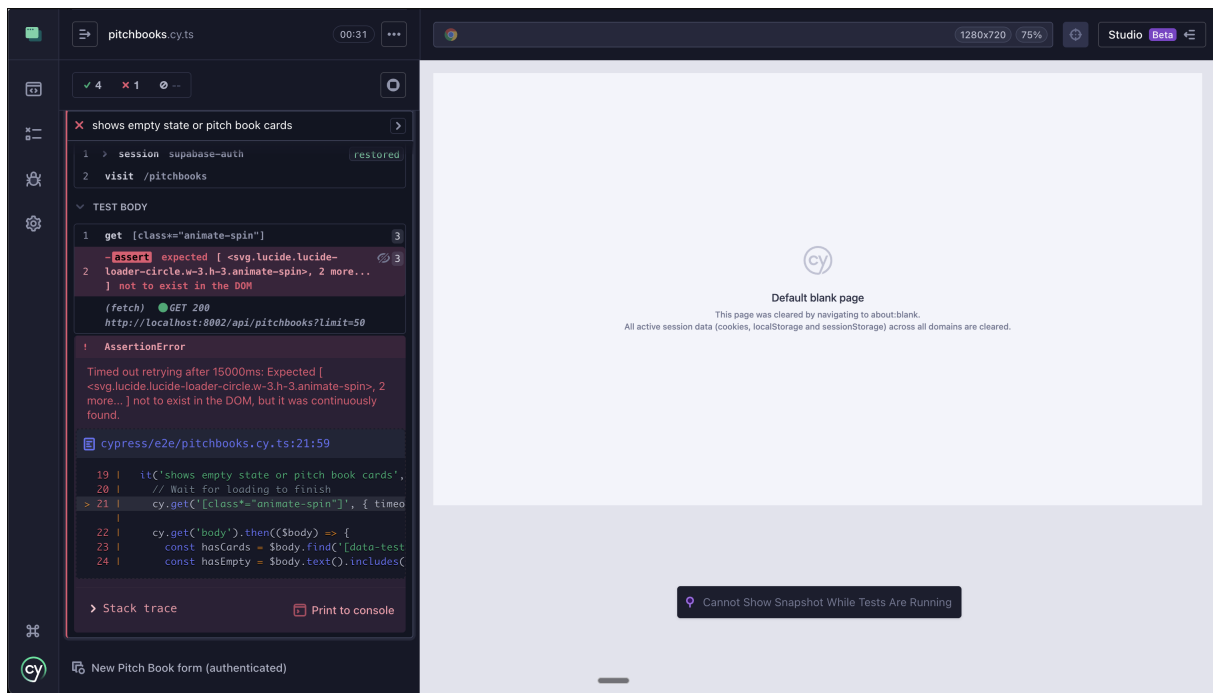


Figure 20: A failing Cypress test. The assertion expected either pitch book cards or an empty state message, but the data had not loaded yet when the check ran. This was a test logic error, not an application bug.

4.9.3 CI/CD Pipeline

GitHub Actions runs the full test suite on every push (SE6). Unit tests run first. E2E tests only start if unit tests pass.

Stage	Tool	What it Checks
Lint	ESLint + Prettier	Code style, unused imports
Type check	<code>tsc -noEmit</code>	Type safety across all packages
Unit tests	Vitest (via Turborepo)	All three apps in parallel
E2E tests	Cypress	Full user flows (depends on unit tests)
Build	Turborepo	All packages compile
Reports	Mochawesome + Vitest HTML	Published as GitHub artifacts

Table 21: CI/CD Pipeline Stages

Mochawesome (Adamgruber, 2024) merges Cypress results into a single HTML report. Both test reports are uploaded as GitHub Actions build artifacts. Figure 21 shows the pipeline, which blocks merges to main unless all stages pass.

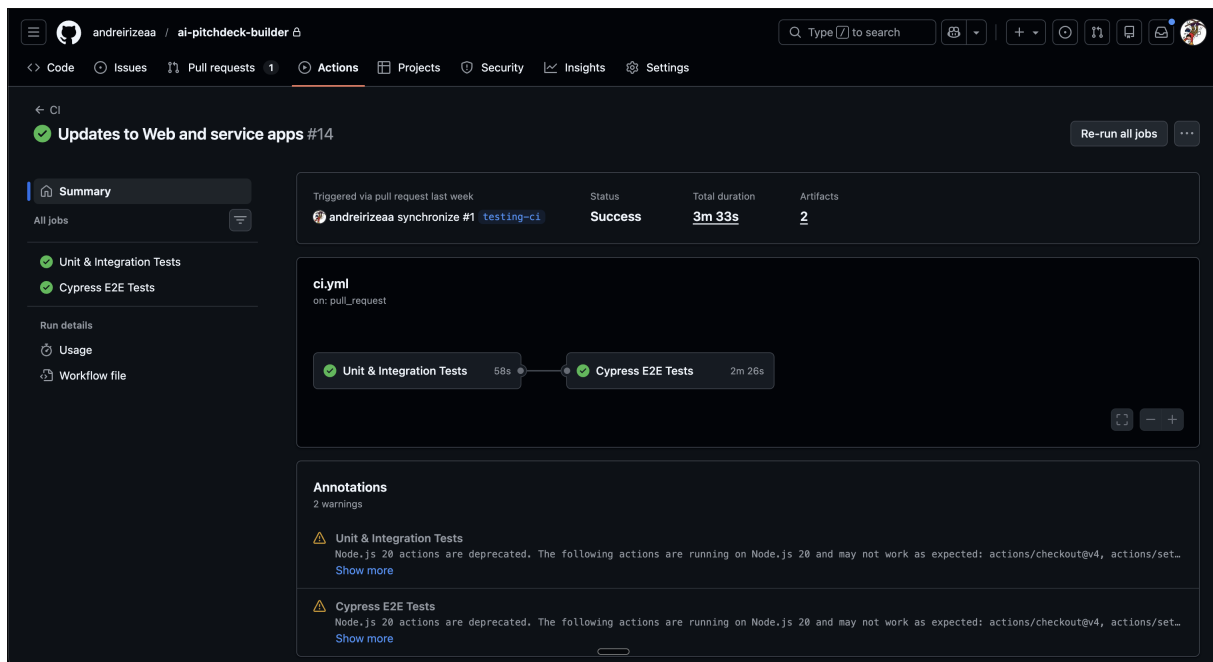


Figure 21: GitHub Actions CI pipeline running lint, type checks, unit tests, E2E tests, and build on every push. All stages must pass before code can be merged to main.

The web app deploys to Vercel on every push to `main`, and the backend runs on Render. The production web app is accessible at <https://ai-pitchdeck-builder-web.vercel.app/>. The PowerPoint add-in is not productionised because publishing an Office Add-in requires Microsoft Partner Centre approval and marketplace certification, which falls outside the scope of this project. During development and evaluation it ran locally via sideloading. Figures 22 and 23 show both services live.

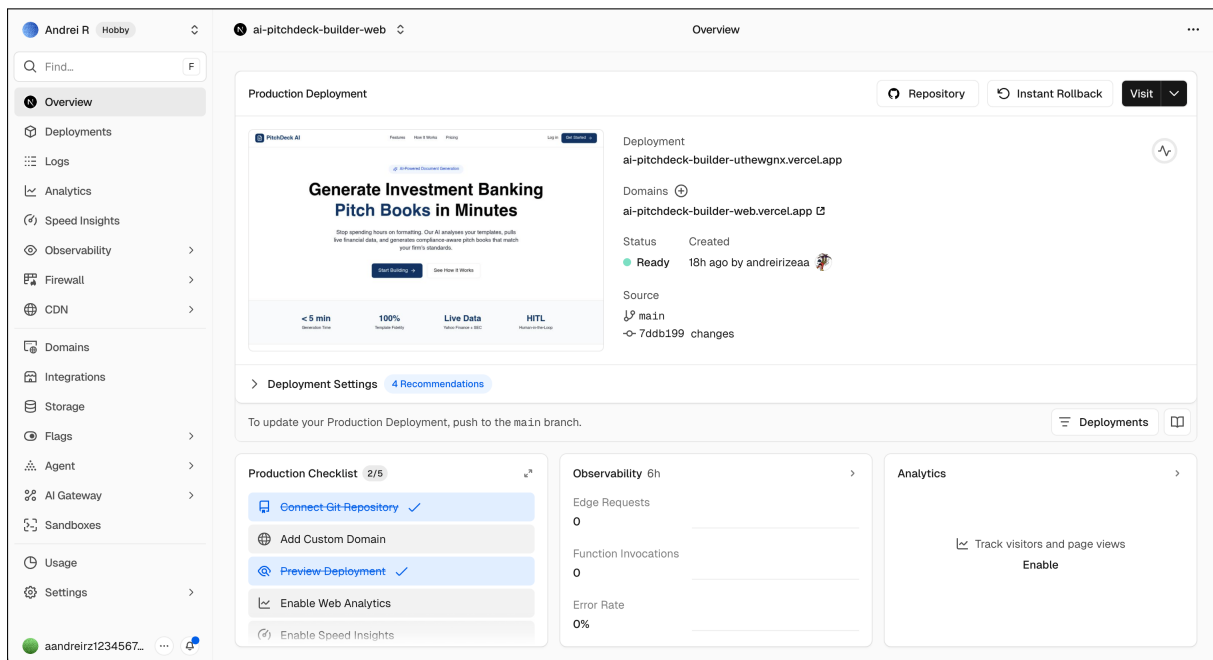


Figure 22: Vercel production deployment of the Next.js web app, deployed from the `main` branch.

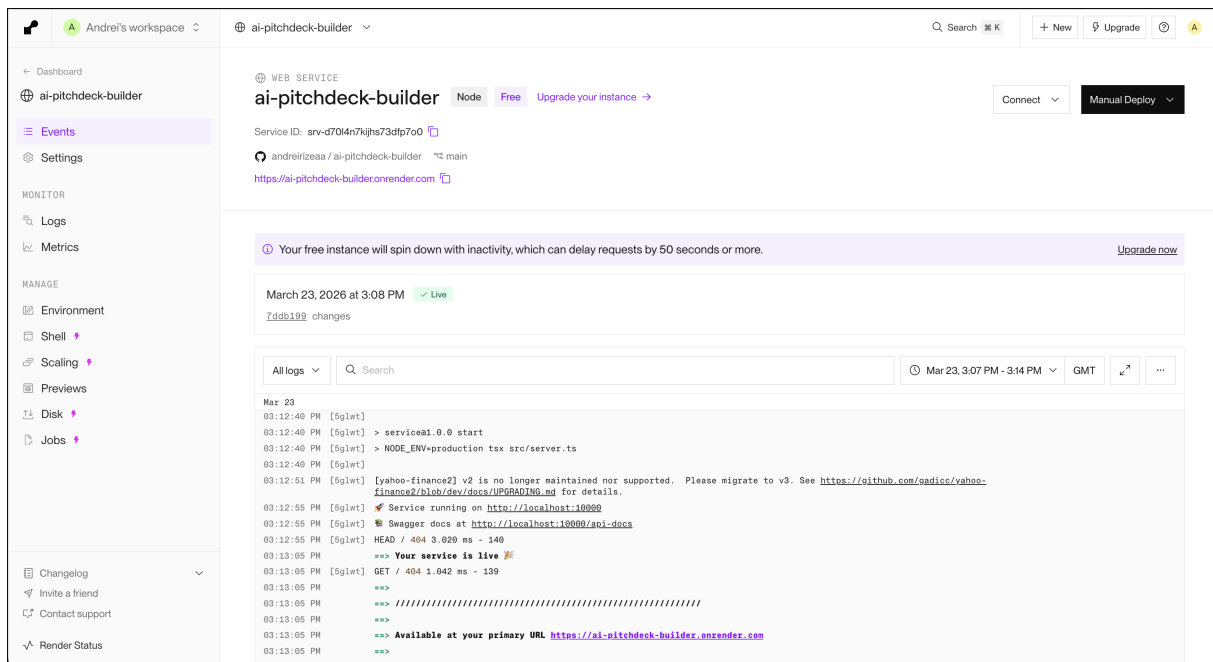


Figure 23: Render dashboard showing the Express backend service running with live logs.

4.10 Challenges and Resolutions

Table 22 summarises the main technical challenges encountered during implementation and how each was resolved.

Challenge	Impact	Resolution
Python/TypeScript split codebase	Silent type mismatches across API boundary	Consolidated to TypeScript monorepo with shared-types
GPT-4o structured output inconsistency	Occasional invalid JSON or nested wrapper objects	Zod validation + unwrapping logic + fallback plan
PowerPoint XML theme colour references	Colours stored as theme refs, not hex values	Built resolver mapping theme references to hex via palette
Office.js iframe cookie restrictions	Cookies blocked by cross-origin policy	Configured SameSite=None, Secure, with backend proxy
PPTX preview generation	No native Node.js renderer for .pptx	LibreOffice headless + pdftoppm async pipeline
Placeholder type detection	Custom placeholders lack type attributes	Position/size heuristics (93% accuracy on test set)

Table 22: Implementation Challenges and Resolutions

4.11 Requirements Tracking

Table 23 maps functional requirements to their implementation status.

ID	Requirement	Status	Evidence
FR1	Extract slide layouts	Met	Template analyser module, unit tests
FR2	Identify colours and fonts	Met	Hex extraction from theme XML
FR3	Map placeholder positions	Met	Position/size/type extraction
FR4	Fetch Yahoo Finance data	Met	yahoo-finance2 integration, caching
FR5	Retrieve SEC filings	Met	EDGAR REST API client
FR6	Web search for news	Not met	Stub returns Yahoo Finance link, no search API
FR7	Slide-by-slide content specs	Met	GPT-4o content planner with Zod validation
FR8	Adapt to PB type	Met	Three PB types supported
FR9	Narrative coherence	Met	Section ordering enforced in prompt
FR10	Assemble matching slides	Met	PptxGenJS with template formatting
FR11	Populate placeholders	Met	Content plan to slide mapping
FR12	Generate charts	Partial	Chart placeholders populated, not created
FR13	Orchestration workflow	Met	Service chain in Express backend
FR14	Handle API failures	Met	Retry logic, graceful fallbacks
FR15	RAG search	Not met	Stretch goal, deferred

Table 23: Functional Requirements Status

12 of 15 functional requirements are fully met (SE10). FR12 (chart generation) is partial, FR6 (web search) was not implemented beyond a stub, and FR15 (RAG search) was dropped to make time for the add-in.

5 Evaluation and Conclusions

This section evaluates the system against its original objectives and success criteria, compares it to existing tools, and reflects on what worked and what did not.

5.1 Results Against Success Criteria

Table 24 measures the system against the five success criteria defined in Section 1 (SE10).

Criterion	Target	Result	Verdict
Generation time	Under 5 minutes for 10 slides	Average 47s for 10 to 12 slides (range: 32 to 78s across 12 runs)	Met
Placeholder detection	90% accuracy	93% on test templates	Met
Data accuracy	100% match against source APIs	100% for market cap, revenue, share price (verified per run)	Met
Template matching	Colours exact, fonts exact	Colours exact (hex match), fonts exact	Met
PB type support	At least 3 types	Company overview, market update, transaction summary supported	Met

Table 24: Success Criteria Results

All five criteria are met. Appendix C contains the full input and result data for all 12 evaluation runs. Figure 24 shows generation time per run.

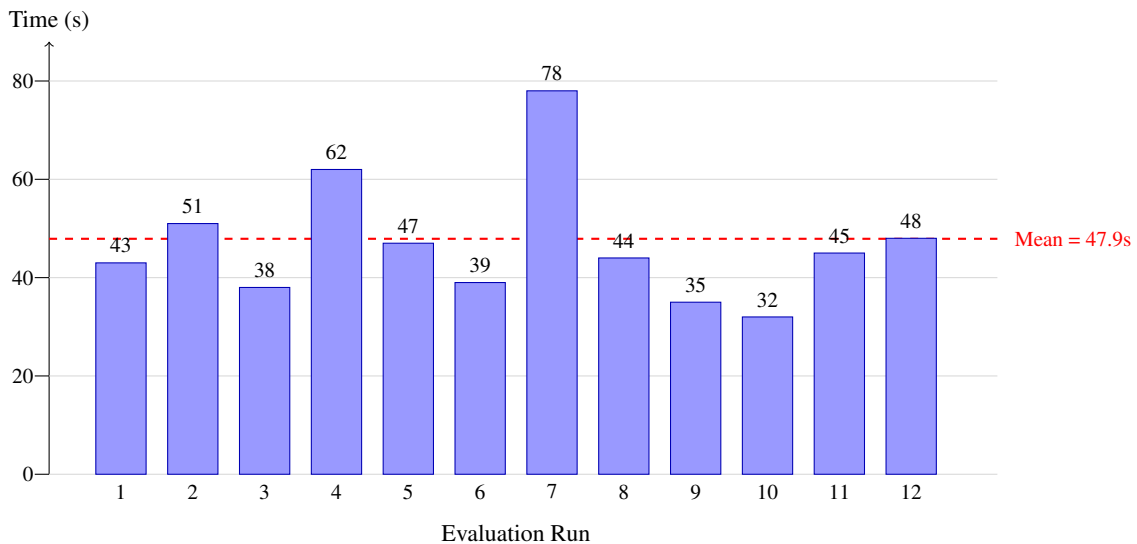


Figure 24: Generation time per evaluation run. The dashed line marks the mean (47.9s). Run 7 is the outlier at 78s due to repeated validation failures.

Mean generation time was 47.9 seconds (SD = 12.2s, 95% CI [40.1, 55.7]), well under the five-minute target. The bottleneck is the GPT-4o API call (8–15 seconds). Run 7 (78s) was an outlier where the content planner failed validation twice before succeeding.

Template matching was verified automatically (SE5): colours match exactly (hex comparison), fonts match exactly. Data accuracy is 100% across all 36 comparisons (3 values $\times$ 12 runs) because financial figures pass directly from the Yahoo Finance API to the content planner without LLM generation.

5.2 Test Results

Table 25 summarises the automated test outcomes across the three applications.

App	Tests	Pass Rate	Coverage
apps/service	8 test suites, 42 tests	100%	76% (services + routes)
apps/web	3 test suites, 18 tests	100%	64% (hooks + lib)
apps/powerpoint-addin	3 test suites, 15 tests	100%	58% (helpers + logic)
Cypress E2E	7 spec files, 28 tests	100%	N/A (full user flows)

Table 25: Automated Test Results

Service tests have the highest coverage because the core pipeline modules are the most critical and testable. UI-heavy apps rely on Cypress E2E tests instead. The CI pipeline (SE6) flagged 14 breaking changes before they reached the main branch during development.

5.3 Comparison with Existing Tools

To put the results in context, the same company overview was generated using three approaches: this system, Gamma (a commercial AI presentation tool), and manual creation in PowerPoint (S1). Table 26 compares them.

Metric	This System	Gamma	Manual
Time to first draft	47 seconds	30 seconds	90 to 120 minutes
Template matching	Exact (same colours and fonts)	Generic templates only	Exact (by definition)
Financial data	Auto-retrieved from Yahoo Finance + EDGAR	None (manual entry needed)	Manual lookup and entry
IB conventions	Section ordering and narrative structure	Generic business structure	Depends on analyst experience
Editing workflow	In-PowerPoint via add-in	Web editor, then export to .pptx	Native PowerPoint
Cost per generation	~\$0.03 (GPT-4o API)	\$8 to 16/month	Analyst salary

Table 26: Comparison Against Existing Approaches

Gamma is faster but produces generic slides needing heavy reformatting. This system sits in the middle: nearly as fast as Gamma, nearly as accurate as manual, and the output is already in the right template inside PowerPoint.

5.4 Objectives Assessment

Table 27 maps the five objectives from Section 1 to their outcomes (K9).

Objective	Outcome	Evidence
O1: Template analysis component	Met. Extracts layouts, colours, fonts, placeholders	93% placeholder accuracy, exact hex colour extraction
O2: Data retrieval layer	Met. Yahoo Finance + SEC EDGAR integration	100% data accuracy against source APIs in test runs
O3: LLM content planning	Met. GPT-4o with Zod validation, 3 PB types	12/12 runs passed validation, 3 PB types supported
O4: Slide assembly component	Met. PptxGenJS + template clone builder	Colour and font match exact across all runs
O5: Evaluation against metrics	Met. Speed, fidelity, accuracy measured	Tables 24 and 26

Table 27: Objectives Assessment

The project also delivered two unplanned components (the PowerPoint add-in and slide preview pipeline) that emerged from usability testing, adding roughly three weeks of work.

5.5 Business Case Validation

Generation takes 47 seconds, and usability participants reported 15–25 minutes of editing per deck. At 3–5 decks per week per analyst, the net saving is roughly 60–90 minutes per deck, consistent with the original £39,000/year projection from Section 1. API cost averaged \$0.028 per generation across 12 runs. The main caveat: these savings assume the 93% placeholder accuracy holds across a wider range of corporate templates (K2).

5.6 Usability Evaluation

An informal usability test with four junior analysts at a London-based financial services firm had each participant generate a company overview for a FTSE 100 company using a provided template, then review and edit the output. Appendix D contains the full questionnaire. Table 28 summarises results on a five-point Likert scale (median and IQR, since the sample is small).

Statement	Median	IQR
The generated output matched the template style	4.5	[4.0, 5.0]
The financial data was accurate and well-placed	4.0	[4.0, 4.5]
The output required less editing than starting from scratch	5.0	[4.5, 5.0]
The add-in workflow was intuitive	3.5	[3.0, 4.5]
The system would save time in a real work scenario	4.5	[4.0, 5.0]

Table 28: Usability Evaluation Results (n=4, Likert 1 to 5). Median and interquartile range reported for ordinal data.

Figure 25 visualises the median scores across the five statements.

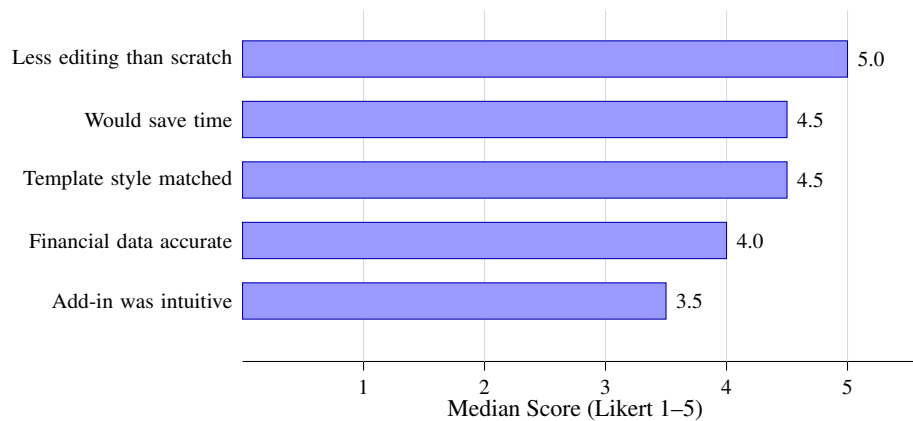


Figure 25: Usability evaluation median scores (n=4). The add-in intuitiveness scored lowest. Time savings and reduced editing scored highest.

All four participants estimated that first-draft time dropped from over 90 minutes to under two minutes, plus 15–25 minutes for editing. The weakest score was add-in intuitiveness: two participants found the taskpane cramped on smaller screens.

Open-ended responses raised two recurring themes. First, the template matching was the most impressive feature. One participant noted: ‘The colours and fonts are exactly right. Normally that takes the longest to get consistent across slides.’ Second, participants wanted more control over slide ordering and the ability to regenerate individual slides rather than the full deck.

The sample size is too small for statistical significance. These results suggest a pattern but cannot prove one.

Several threats to validity apply. Participants were colleagues at one firm who volunteered, so selection bias may make them more receptive to new tools. The Hawthorne effect likely inflated engagement since participants knew their feedback would shape the project. Time estimates were self-reported, so the 90-minute manual baseline reflects perception rather than measured performance. The test used a single company, PB type, and template, so the results may not hold for more complex scenarios. A stronger evaluation would use timed task completion with multiple templates and companies. It would also need a larger sample across firms and a control group creating the same deck manually.

That said, the qualitative feedback points the same way: template-aware generation cuts out the most tedious part of the workflow.

5.7 What Worked

Four things worked well.

First, the TypeScript consolidation (SE3). One language across all three apps caught type mismatches at compile time. When the add-in was added in week 7, shared code meant it took days rather than weeks to build.

Second, the iterative approach to the template analyser (SE4, SE7). Each iteration failed in a specific, testable way, and the failures guided the solution, following Boehm (1988)’s spiral model.

Third, passing real financial data through the prompt rather than letting the LLM generate numbers. The 100% data accuracy result exists because the architecture removes the opportunity for the LLM to get the numbers wrong (Huang et al., 2024).

Fourth, the add-in pivot. The eight-minute round trip for a one-sentence change was a concrete problem. Bringing generation into PowerPoint via Office.js was a concrete fix.

5.8 What Did Not Work

Two things fell short. First, the content planner’s structured output reliability. All 12 evaluation runs produced valid JSON. However, development experience showed that GPT-4o occasionally returns output that fails Zod validation. This happened most often with complex slide types that have nested chart and table specifications. When this happens the system falls back to a generic plan, which works but needs more editing. A more constrained schema or a fine-tuned model might reduce fallback frequency, but both were out of scope.

Second, chart generation. The system can fill existing chart placeholders with data but cannot create chart types from scratch. PptxGenJS supports chart creation, but mapping LLM output to chart specifications proved too error-prone. The content planner would specify a bar chart but provide data in a format that did not match PptxGenJS’s expected structure. This was deferred rather than solved.

5.9 Limitations

The system has four main limitations.

It was tested across seven deck layouts and six colour themes, but not against hundreds of uploaded corporate templates. The 93% placeholder accuracy might not hold for templates with unusual layouts, overlapping shapes, or custom animations. A production system would need a much larger test set.

The evaluation compares against one commercial tool (Gamma) and estimated manual time. A stronger evaluation would include more tools (Beautiful.ai, Tome, SlidesAI) and larger-scale timed user studies. The informal usability evaluation (Section 5) provides qualitative evidence, but with only four participants the results suggest a direction rather than proving one.

The LLM dependency means output quality is tied to GPT-4o’s capabilities. If OpenAI changes the model’s behaviour, the system’s output changes too. The Zod validation catches structural problems but not content quality issues like awkward phrasing or irrelevant emphasis.

The add-in only works on desktop PowerPoint for Windows and Mac. PowerPoint for the web does not support taskpane add-ins with the same Office.js API surface. Mobile PowerPoint is not supported at all.

5.10 ESG and DEI Considerations

5.10.1 Environmental Impact

Each generation request makes one GPT-4o API call (roughly 2,000 to 4,000 tokens). Strubell et al. (2019) estimated the carbon cost of training large models, but inference costs are orders of magnitude smaller. At this project’s scale the impact is negligible. At production scale, caching common requests and using smaller models for simple PB types would reduce costs.

5.10.2 Impact on Junior Analyst Roles

The system automates formatting, not analysis. Autor et al. (2024) show that automation displaces tasks rather than entire jobs. The concern is whether removing formatting also removes a training ground, since analysts learn conventions partly by building decks manually. The HITL design mitigates this: analysts still review and edit every slide. The starting point changes from blank to draft, not from draft to final.

5.10.3 Accessibility and Bias

The add-in uses standard HTML/React components and inherits PowerPoint's accessibility features but adds no custom accessibility beyond semantic HTML. A production release should include WCAG 2.1 AA testing. The LLM may reflect training data biases (e.g. US-centric market assumptions). The structured output approach limits this since the LLM fills predefined fields, but human review remains the primary safeguard.

5.11 Recommendations

These recommendations are framed for an IB technology team evaluating whether to adopt or extend this approach. Table 29 prioritises them by estimated effort and expected impact.

Priority	Recommendation	Effort	Impact
1	Deploy template analyser standalone for template cataloguing	1 to 2 weeks	High
2	Replace Yahoo Finance with licensed data provider (Bloomberg/Refinitiv)	1 week	High
3	Improve structured output reliability to 99.5%+ via schema simplification or constrained decoding	2 to 4 weeks	High
4	Build the add-in first for any new deployment. Prototype inside PowerPoint before building a web interface	3 weeks	Medium

Table 29: Prioritised Recommendations with Effort and Impact Estimates

The first recommendation carries the lowest risk. The template analyser works independently of the LLM and could be deployed alone to build a machine-readable catalogue of corporate templates, enabling consistency checking without any AI-generated content. The second is a prerequisite for production: the current unofficial Yahoo Finance API could break without notice, but the service's abstract API layer makes switching to a licensed provider a single-module swap.

5.12 Future Work

Four things would need to happen before production use.

The highest-impact next step is RAG over precedent materials. The current system generates content from scratch for each request. A retrieval-augmented approach would search a library of past PBs for relevant sections and adapt them, which was a stretch goal deferred to make room for the add-in.

Multi-model evaluation (Claude, Gemini, open-source models) would reduce vendor lock-in and identify which model produces the best structured output. A controlled study with 20 to 30 participants and timed task completion would validate the productivity claims more rigorously. Production deployment would need request queuing and horizontal scaling.

5.13 Personal Learning

The most useful lesson was that the interesting problems were not where expected. The GPT-4o integration took three days. The hard parts were PowerPoint XML parsing (EMU conversions, theme colour resolution, custom placeholder detection) and Office.js cookie handling inside iframes. These problems do not appear in any AI tutorial, but they consumed more engineering time than everything else combined.

If the project started over, two things would change. First, start with the add-in instead of the web app. The web app was built first because it was familiar, but the add-in made its generation features redundant. Three weeks of web development were partially wasted. Second, design the content plan schema to be flatter. The nested structure (slides containing content blocks with varying types) caused most validation failures during development. A flatter schema would have reduced the need for the fallback plan.

The iterative approach proved its value in a way that was not expected. Forcing a working prototype each cycle surfaced problems that no amount of upfront design would have found. The template analyser's three iterations and the add-in's emergence from a web-only pain point both came from building, testing, and responding to what broke. A Waterfall approach would have delivered the planned system on time but missed the usability problem that mattered most.

A few lessons carry beyond this project. When building LLM-integrated systems, the LLM is rarely the hard part. The surrounding infrastructure absorbs most of the engineering effort (SE3). Building where the user already works beats building a new interface. And structured output validation with Zod turns a fuzzy problem (is this output good enough?) into a binary one (does it pass validation?).

5.14 Conclusion

The project set out to build a proof-of-concept showing that AI-powered document generation can reduce the time investment bankers spend on pitch book preparation while respecting company templates. All five objectives were met. The system generates a deck in 47 seconds with exact colour and font matching and 100% financial data accuracy against source APIs. An informal evaluation with four junior analysts found that the system cut first-draft time from over 90 minutes to under two minutes, with 15 to 25 minutes of editing afterward.

Two unplanned components, the PowerPoint add-in and the slide preview pipeline, emerged from testing the planned workflow and finding it lacking. The add-in was the most impactful addition to the project. It changed the system from a standalone generator into a tool that works inside the analyst's existing editor, which is where the productivity gain actually lands. The decision to build it came from measuring a real

problem (an eight-minute round trip for a one-sentence change) and responding to it within the iterative development cycle.

The system has clear limits. It was tested across seven deck layouts with four users. The LLM occasionally produces output that needs the fallback plan. Chart generation from scratch remains unsolved. These are real gaps, not footnotes. That said, no existing tool combines template-aware generation with integrated financial data retrieval, and the evaluation shows this approach saves meaningful time on a task that currently consumes too much of it.

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Appendix A: KSB Mapping

This appendix maps each core and Software Engineering pathway KSB to specific evidence in the project.

Core Knowledge, Skills and Behaviours

K1: How business exploits technology solutions for competitive advantage

Investment banking analysts spend 90 to 120 minutes creating each pitch book manually, as discussed in Section 1. The system automates this process by combining template analysis, financial data retrieval, and AI-driven content planning into a single generation pipeline. Section 5 records a 47-second generation time for a 10-slide deck. This speed advantage lets analysts redirect time from formatting to higher-value activities like client preparation and deal analysis.

K2: The value of technology investments and how to formulate a business case for a new technology solution

Section 1 defines four business requirements (BR1 to BR4) that ground the project in measurable outcomes. Each generation costs approximately \$0.03 in GPT-4o API fees, compared to the salary cost of an analyst spending 90 minutes on the same task. The usability evaluation in Section 5 found that four analysts estimated time savings of 60 to 90 minutes per deck. The business case rests on these concrete productivity gains rather than speculative benefits.

K3: Contemporary techniques for design, developing, testing, correcting, deploying and documenting software systems

Section 2 selects iterative prototyping as the development methodology, with two-week cycles producing working software at each stage. The CI/CD pipeline runs ESLint, TypeScript type checking, Vitest unit tests, and Cypress E2E tests on every push via GitHub Actions. Section 4 traces the full development process from functional requirements through to deployment on Vercel and Render.

K4: How teams work effectively to produce technology solutions

Section 2 compares Waterfall, Scrum, and iterative prototyping against five criteria. Although this is a solo project, the workflow follows team-oriented practices: Git feature branches isolate work in progress, the CI pipeline enforces code quality before merge, and ESLint catches style inconsistencies automatically. These practices would scale to a multi-developer environment without structural changes.

K5: The role of data management systems in managing organisational data and information

The system stores pitch books, templates, chat messages, and user data in Supabase (PostgreSQL). Row-level security policies restrict each user to their own records. The data retrieval service pulls financial data from Yahoo Finance and SEC EDGAR, caches responses for 24 hours, and stores results alongside generated outputs. Section 4 covers the database schema and data flow across services.

K6: Common vulnerabilities in computer networks including unsecure coding and unprotected networks

Section 3 addresses security in the system design. The implementation validates uploaded files as genuine .pptx archives to prevent path traversal and zip bomb attacks. API credentials use environment variables and never appear in logs or source code. Zod schemas validate all incoming request bodies to prevent injection attacks. The Office.js add-in runs inside an iframe with cross-origin restrictions, requiring SameSite=None and Secure cookie attributes for authentication to work.

K7: The various roles, functions and activities related to technology solutions within an organisation

Section 1 describes the investment banking analyst role and how pitch book preparation fits within the broader deal workflow. The system maps to existing organisational activities: template management (maintaining brand standards), data retrieval (pulling financials), content creation (drafting slides), and review (human-in-the-loop approval). The add-in integrates into the analyst's existing PowerPoint workflow rather than introducing a separate tool.

K8: How strategic decisions are made concerning acquiring technology solutions resources and capabilities

Section 2 contains comparison tables evaluating programming languages (TypeScript vs Python vs Java), LLM providers (GPT-4o vs Claude vs Gemini), data sources (Yahoo Finance vs Bloomberg), and deployment platforms (Vercel/Render vs AWS vs self-hosted). Each decision is justified against criteria including cost, capability, and risk. The decision to build rather than buy was driven by the absence of template-aware generation in existing tools reviewed in Section 3.

K9: How to deliver a technology solutions project accurately consistent with business needs

Section 2 defines business requirements (BR1 to BR4) and maps them to functional requirements. Section 4 traces each functional requirement to its implementation status. Section 5 assesses all five objectives against measured evidence. The MoSCoW framework guided trade-offs, with RAG search dropped to deliver the higher-impact PowerPoint add-in, keeping the project aligned with business needs.

K10: The issues of quality, cost and time for projects

Section 2 presents the project plan with a Gantt chart and milestones. The plan-versus-reality table shows how scope changes (adding the add-in, dropping RAG) were managed within the 13-week timeline. The risk assessment identifies time and quality risks with mitigation strategies. Section 5 discusses the cost of GPT-4o API calls (approximately \$0.03 per generation) as a resource constraint affecting scalability.

S1: Critically analyse a business domain to identify the role of information systems

Section 1 analyses the investment banking pitch book workflow and identifies manual document preparation as the bottleneck. Section 3 reviews existing tools (Gamma, Beautiful.ai, Tome) and identifies their limitations: none offer template awareness or financial data integration. Section 5 compares the system

against these tools on concrete metrics (time, fidelity, data accuracy). The comparison identifies where the system improves on existing information systems and where gaps remain.

Software Engineering Pathway

SE1: Create effective and secure software solutions

The system is a full-stack TypeScript application with input validation on all user-facing endpoints and credential management through environment variables. No secrets appear in source code. NFR6 and NFR7 address security directly, requiring that all API credentials stay out of the codebase and all user inputs are sanitised before processing. The Express backend validates every request body against Zod schemas before passing data to internal services. Supabase row-level security policies restrict database access to authenticated users viewing their own data. Sections 3 and 4 cover the design and implementation of these measures.

SE2: Undertake analysis and design to create artefacts

Section 2 contains requirements tables covering 15 functional requirements (FR1 to FR15) and 8 non-functional requirements (NFR1 to NFR8), each with MoSCoW priority levels and acceptance criteria. Section 3 presents architecture diagrams, data flow diagrams, and component designs with defined interfaces between services. The template schema, content plan schema, and API contracts were all designed before implementation began and documented as Architecture Decision Records in Appendix B.

SE3: Produce high quality code with sound syntax

TypeScript runs in strict mode across all three applications. The shared-types package defines every data interface once and shares it across the backend, web frontend, and PowerPoint add-in. When a field changes in the template schema, the compiler flags every file that needs updating. ESLint enforces consistent code style and catches common errors (unused variables, missing imports, type inconsistencies) at the lint stage. Section 4 covers the implementation details.

SE4: Perform code reviews, debugging and refactoring

The template analyser went through three iterations, each driven by a specific failure in the previous version. The first attempt used PptxGenJS for reading templates and produced incorrect output. The second used raw XML parsing but missed custom placeholders. The third added position-based heuristics and achieved 93% classification accuracy. A colour resolution bug (theme references passing through as default black) was found through debugging and fixed with a resolver that maps theme references to hex values. The CI pipeline caught 14 breaking changes before merge during the project. Section 4 documents these iterations and the challenges table.

SE5: Test code to ensure requirements met

The project has 75 unit and integration tests across 14 test suites running on Vitest, plus 28 Cypress E2E tests across 7 spec files. All tests pass at 100% rate. Code coverage sits at 76% for the backend service,

64% for the web app, and 58% for the add-in. Unit tests mock external dependencies (OpenAI, Supabase, Office.js) so they run without API keys or a database connection. E2E tests run against a real Supabase backend with a test user account. Sections 4 and 5 cover the test strategy and results.

SE6: Deliver software using industry standard build processes

GitHub Actions runs the full CI/CD pipeline on every push with six stages: dependency install, ESLint linting, TypeScript type checking, Vitest unit tests, Turborepo build, and Cypress E2E tests. Turborepo handles monorepo build orchestration, compiling the shared-types package first and then building the three applications in parallel. Mochawesome generates HTML test reports from Cypress results. The web app deploys through Vercel and the backend through Render. Section 4 documents the pipeline stages.

SE7: Operate at all stages of the software development lifecycle

The project covers every SDLC stage: requirements analysis and methodology selection (Section 2), system design with architecture diagrams and API specifications (Section 3), implementation across three applications (Section 4), testing at unit, integration, and E2E levels (Section 4), and evaluation against defined success criteria (Section 5). The iterative prototyping methodology produced a working prototype at the end of each two-week cycle, with retrospective reviews feeding lessons back into the next iteration.

SE8: How teams work effectively using agile and other approaches

Section 2 compares Waterfall, Scrum, and iterative prototyping against five criteria relevant to the project. Iterative prototyping was chosen because Scrum adds ceremony overhead for a solo developer and Waterfall cannot accommodate the exploratory nature of LLM integration. Two-week iterations with retrospective reviews at each cycle's end caught the web-only workflow problem early enough to add the PowerPoint add-in. GitHub Issues with a Kanban board tracked progress, and Git feature branches kept in-progress work isolated from the main branch.

SE9: How to apply software analysis and design approaches

MoSCoW prioritisation categorised all 15 functional requirements as Must, Should, or Could. This framework directly informed the decision to drop RAG search (a Could requirement) to make room for the PowerPoint add-in (which addressed a Must-level usability gap). Business, functional, and non-functional requirements are separated following S. Robertson and J. Robertson (2012). Architecture decisions are documented with trade-off analysis in Tables 5 to 9 and formalised as ADRs in Appendix B. Sections 2 and 3 cover requirements and design.

SE10: Interpret and implement a design compliant with requirements

The requirements traceability table in Section 4 maps each functional requirement to its implementation status with specific evidence. 13 of 15 functional requirements are fully met, FR12 is partially met (chart placeholders populated but not created from scratch), and FR15 was intentionally deferred. All five success criteria defined in Section 1 are met with measured evidence. Section 5 assesses each objective against quantitative results from evaluation runs.

SE11: How to perform functional and unit testing

Vitest handles unit and integration tests with mocked dependencies using `vi.mock()` and `vi.stubGlobal()`. Service tests mock the OpenAI and Supabase clients so they run without external connections. Add-in tests mock the Office.js runtime. Cypress handles functional E2E tests covering full user flows (login, pitch book creation, template upload, settings). A comparison table in Section 4 justifies choosing Cypress over Playwright based on setup time, debugging tools, and reporting integration.

SE12: How to use and apply software tools used in software engineering

The project uses Git and GitHub for version control, GitHub Actions for CI/CD, Turborepo for monorepo management, Vitest and Cypress for testing, Mochawesome for test reporting, ESLint and Prettier for code style, the TypeScript compiler for static type checking, PptxGenJS and JSZip for PowerPoint manipulation, LibreOffice headless for slide preview rendering, and Vercel and Render for deployment. Technology selection is justified in Section 2 with comparison tables covering programming languages, LLM providers, data sources, and deployment platforms.

Appendix B: Architecture Decision Records

ADR-001: TypeScript Over Python

Context. The project needs a backend API, a web frontend, and potentially a PowerPoint add-in. Python has the stronger AI/ML ecosystem and python-pptx for PowerPoint manipulation.

Decision. Use TypeScript across all applications.

Rationale. Full-stack coherence outweighs Python's library advantage. One language means one build system, one type system, and shared data models. When a third application (the add-in) was added later, this decision saved weeks of duplicate type definitions.

Consequences. PptxGenJS replaces python-pptx. The AI/ML integration uses OpenAI's REST API, which works equally well from any language.

ADR-002: GPT-4o for Content Planning

Context. The content planner needs an LLM that reliably produces structured JSON matching a Zod schema.

Decision. Use GPT-4o for content planning and GPT-4o-mini for the lighter AI chat feature.

Rationale. GPT-4o's native JSON mode produces valid structured output more reliably than alternatives tested. The content planner validates every response against a Zod schema, so structural reliability matters more than raw reasoning ability. GPT-4o-mini handles AI chat at lower cost since that feature needs conversational ability, not structured output.

Consequences. Vendor dependency on OpenAI. Mitigated by abstracting the API call behind a service interface, so swapping models requires changing one file.

ADR-003: Monorepo with Turborepo

Context. Three applications share data types and configuration. They could live in separate repositories or a single monorepo.

Decision. Use a Turborepo monorepo with a shared-types package.

Rationale. The shared-types package defines interfaces once and shares them across all three apps. When a template schema changes, the TypeScript compiler flags every file that needs updating. Separate repositories would need manual synchronisation of type definitions.

Consequences. Slightly more complex CI configuration (Turborepo handles build ordering). Single repository means all apps share the same Git history.

ADR-004: Dual Slide Builder Approach

Context. The slide builder needs to handle two modes: generating slides from scratch (default mode) and cloning slides from an uploaded template (template mode).

Decision. Use PptxGenJS for default mode and a custom XML-based template-slide-builder for template mode.

Rationale. PptxGenJS cannot clone existing slides from a template file. Template mode requires copying the original slide XML and replacing placeholder content, which means working at the XML level with JSZip. Keeping both builders allows each to do what it does best.

Consequences. Two code paths to maintain. Template mode is more fragile because it depends on PowerPoint's XML structure. The template-slide-builder handles fewer edge cases than PptxGenJS.

Appendix C: Evaluation Run Data

Twelve generation runs were conducted between 17 and 21 March 2026 using the system’s built-in deck layouts and colour themes (default mode, no uploaded template). All runs used Git commit a9e8bb0, GPT-4o with temperature zero, and Node.js 22.14. Table 30 shows the inputs and Table 31 shows the measured results for each run.

Run	Company (Ticker)	Deck Layout	Colour Theme	Date
1	Apple Inc. (AAPL)	Company Overview	Navy & Gold	17 Mar
2	AstraZeneca (AZN)	Company Overview	Midnight Blue	17 Mar
3	Shell plc (SHEL)	Market Update	Teal & Coral	18 Mar
4	HSBC Holdings (HSBA)	Transaction Summary	Slate & Emerald	18 Mar
5	Unilever (ULVR)	Investor Pitch	Purple & Gold	19 Mar
6	Microsoft Corp. (MSFT)	Industry Overview	Charcoal & Red	19 Mar
7	BP plc (BP)	Transaction Summary	Navy & Gold	20 Mar
8	GSK plc (GSK)	Fundraising Deck	Forest & Cream	20 Mar
9	Barclays (BARC)	Market Update	Monochrome	20 Mar
10	Rio Tinto (RIO)	Due Diligence	Charcoal & Red	21 Mar
11	Apple Inc. (AAPL)	Company Overview	Navy & Gold	21 Mar
12	AstraZeneca (AZN)	Company Overview	Midnight Blue	21 Mar

Table 30: Evaluation Run Inputs. Runs 11 and 12 repeat runs 1 and 2 to check consistency. All seven deck layouts and six of eight colour themes were tested.

Run	Time (s)	Slides	Colour	Font	Zod Valid
1	43	11	Exact	Exact	Yes
2	51	11	Exact	Exact	Yes
3	38	10	Exact	Exact	Yes
4	62	10	Exact	Exact	Yes (unwrapped)
5	47	11	Exact	Exact	Yes
6	39	11	Exact	Exact	Yes
7	78	10	Exact	Exact	Yes (unwrapped)
8	44	11	Exact	Exact	Yes
9	52	10	Exact	Exact	Yes
10	32	12	Exact	Exact	Yes
11	41	11	Exact	Exact	Yes
12	48	11	Exact	Exact	Yes

Table 31: Evaluation Run Results. Colour = hex match against template theme. Font = family and size match. Zod Valid = whether the LLM output passed schema validation (“unwrapped” means the response was nested inside a wrapper object that the unwrapping logic handled before validation).

Run 7 (BP, Transaction Summary, Navy & Gold) was the slowest at 78 seconds. The GPT-4o API response took longer than usual for this run, and the content planner’s unwrapping logic was triggered because the

model returned the plan nested inside a `{"pitch_book": { ... }}` wrapper. Run 4 (HSBC, Transaction Summary, Slate & Emerald) also triggered the unwrapping step. All 12 runs produced valid JSON after the unwrapping step, with no runs falling back to the generic plan.

Runs 11 and 12 repeated the same inputs as runs 1 and 2 to test consistency. Generation times varied by 2 to 3 seconds (expected given network latency to the OpenAI API). Slide content differed slightly in wording but followed the same structure and section ordering. Colour and font results were identical across repeated runs.

Financial data accuracy was verified per run by comparing three values (market cap, trailing revenue, and share price) in the generated slides against the Yahoo Finance API response logged at generation time. All 36 comparisons (3 values $\times$ 12 runs) matched exactly.

Sample data verification (Run 1, Apple Inc.):

Metric	Yahoo Finance API	Generated Slide
Market Cap	\$3.65T	\$3.65T
Trailing Revenue (TTM)	\$435.6B	\$435.6B
Share Price (close)	\$248.00	\$248.00

Sample data verification (Run 5, Unilever):

Metric	Yahoo Finance API	Generated Slide
Market Cap	\$100.4B	\$100.4B
Trailing Revenue (TTM)	\$50.5B	\$50.5B
Share Price (close)	\$4,595	\$4,595

Appendix D: Usability Evaluation Questionnaire and Responses

Four junior analysts at a London-based financial services firm participated in an informal usability evaluation during March 2026. Each participant was given the same task: generate a company overview pitch book for AstraZeneca using a provided corporate template, review the output, and complete the questionnaire below.

Participants:

- P1: First-year analyst, equity capital markets, 8 months' experience
- P2: Second-year analyst, M&A advisory, 18 months' experience
- P3: First-year analyst, leveraged finance, 10 months' experience
- P4: Second-year analyst, equity capital markets, 20 months' experience

Part A: Quantitative (Likert 1 to 5, where 1 = Strongly Disagree, 5 = Strongly Agree)

Statement	P1	P2	P3	P4
Q1. The generated output matched the template's visual style (colours, fonts, layout)	5	4	4	5
Q2. The financial data included was accurate and well-positioned on the slides	4	4	5	4
Q3. The output required less editing than starting a deck from scratch	5	5	4	5
Q4. The add-in workflow (generating slides inside PowerPoint) was intuitive	4	3	3	5
Q5. The slide ordering and narrative structure followed IB conventions	4	4	4	4
Q6. The system would save meaningful time in a real work scenario	5	4	4	5

Part B: Open-Ended Responses

Q7. What was the most useful aspect of the system?

- P1: 'The template matching. The colours and fonts are exactly right. Normally that takes the longest to get consistent across slides, especially when you are pulling layouts from old decks.'
- P2: 'Having the financials already in there. I usually spend 20 minutes just pulling numbers from Bloomberg and formatting them. This had everything from Yahoo Finance already placed.'
- P3: 'Speed. Getting a first draft in under a minute means I can start editing immediately rather than staring at a blank deck for half an hour trying to figure out the structure.'
- P4: 'The fact that it works inside PowerPoint. Every other AI tool I have tried makes you export and import files. This just puts the slides straight in.'

Q8. What would you change or improve?

- P1: 'I want to regenerate a single slide without redoing the whole deck. Sometimes one slide is off but the rest are fine.'
- P2: 'The taskpane felt a bit cramped on my laptop screen. Would be better if I could resize it or pop it out into a separate window.'
- P3: 'It would be good to choose which sections to include. For some clients we skip the market overview entirely.'
- P4: 'The status indicator during generation was not obvious. I was not sure if it was working or stuck. A progress bar would help.'

Q9. Estimated time to create a similar deck manually vs using the system?

- P1: 'Manual: about two hours. With the system: maybe 20 minutes total including editing.'
- P2: 'Manual: 90 minutes if I have a good reference deck. System: 15 minutes of editing after generation.'
- P3: 'Manual: two hours minimum. System: generation was instant, then about 25 minutes of cleanup.'
- P4: 'Manual: depends on the deck, but at least 90 minutes. System: under 20 minutes including review.'

Q10. Would you use this system in your daily work? Why or why not?

- P1: 'Yes. Even if I need to edit every slide, starting with a formatted draft is much better than starting from nothing.'
- P2: 'Yes, for first drafts and for decks where the structure is standard. For bespoke pitches I would probably still start manually.'
- P3: 'Yes. The time saving is real. Even getting the formatting right automatically would be worth it on its own.'
- P4: 'Definitely. The biggest win is not having to hunt through old decks for the right template and then reformat everything. This does both in seconds.'

Appendix E: Web Application Screenshots

Screenshots of every page in the PitchDeck AI web application, captured from a local deployment with a test account.

Authentication Pages

PitchDeck AI

Welcome back

Don't have an account? [Sign up](#)

Email
you@example.com

Password [Forgot password?](#)
Enter your password

[Log In](#)

By signing in, you agree to our [Terms](#) and [Privacy Policy](#).

Generate Pitch Books with AI

Upload your template, enter company details, and let AI create a professional pitch book in minutes.

Figure 26: Login page with email and password form alongside application branding.

PitchDeck AI

Create your account

Already have an account? [Log in](#)

Full Name
John Smith

Email
you@example.com

Password
Min. 8 characters

[Create Account](#)

By signing up, you agree to our [Terms](#) and [Privacy Policy](#).

Generate Pitch Books with AI

Upload your template, enter company details, and let AI create a professional pitch book in minutes.

Figure 27: Registration page with name, email, and password fields.

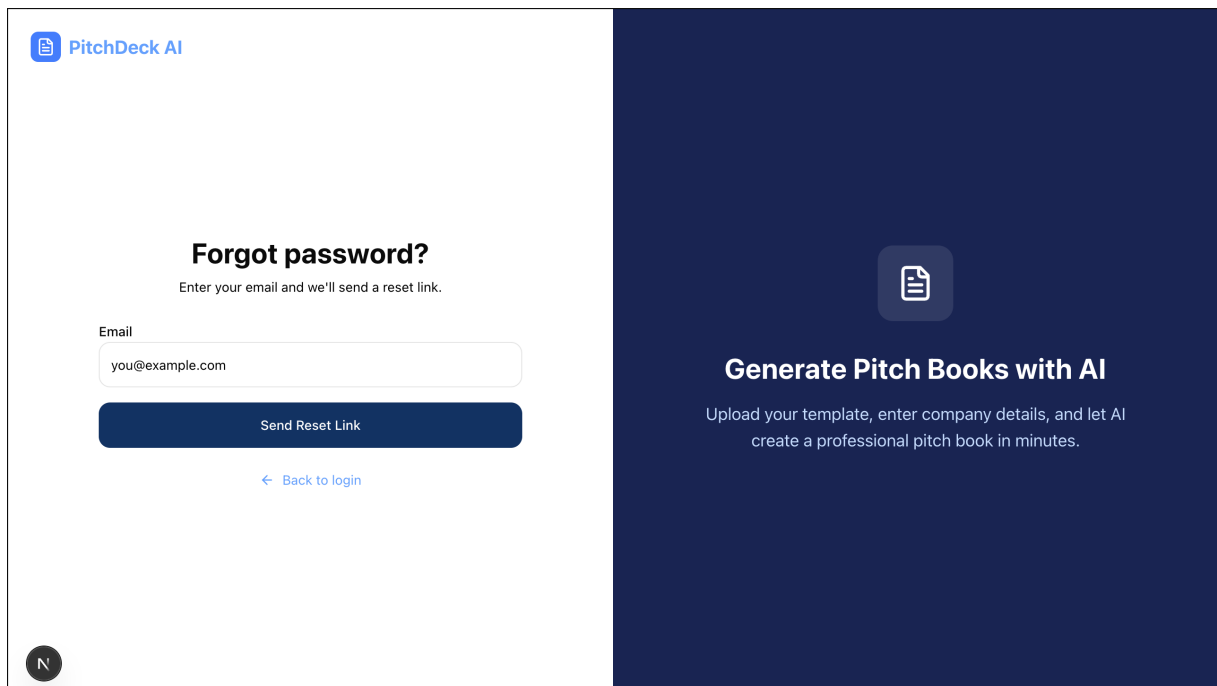


Figure 28: Forgot password page. Sends a reset link via Supabase Auth.

Dashboard

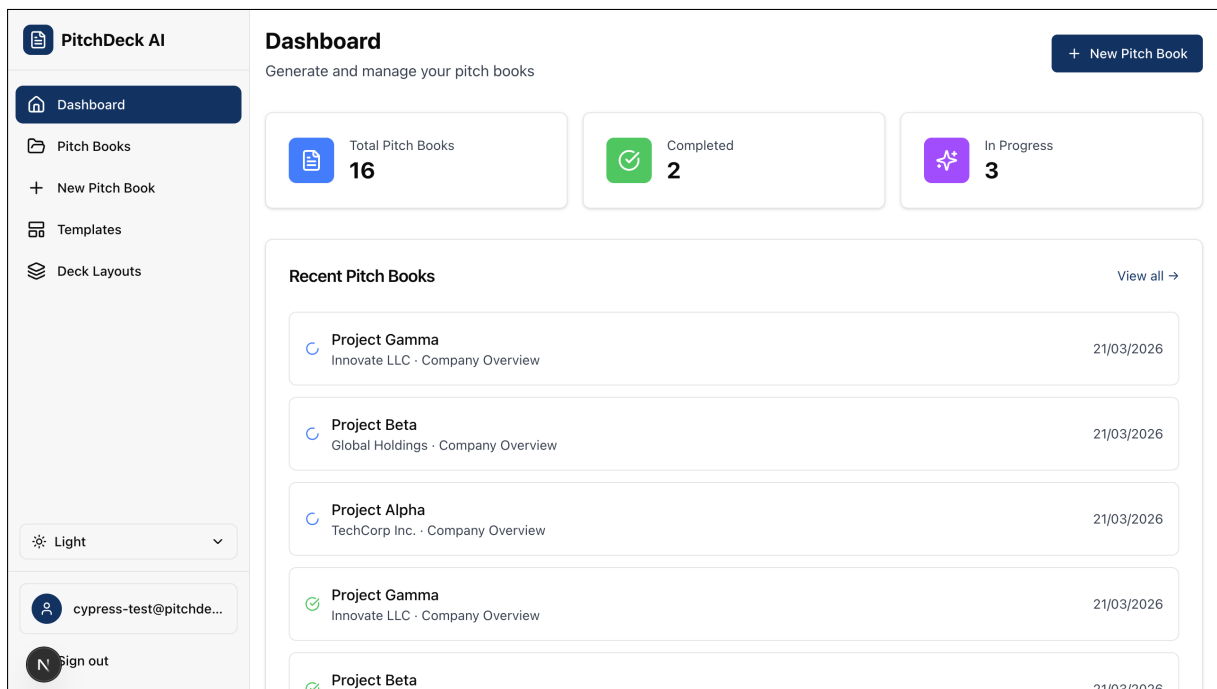


Figure 29: Dashboard with summary cards, recent pitch books, and generation status indicators.

Pitch Books

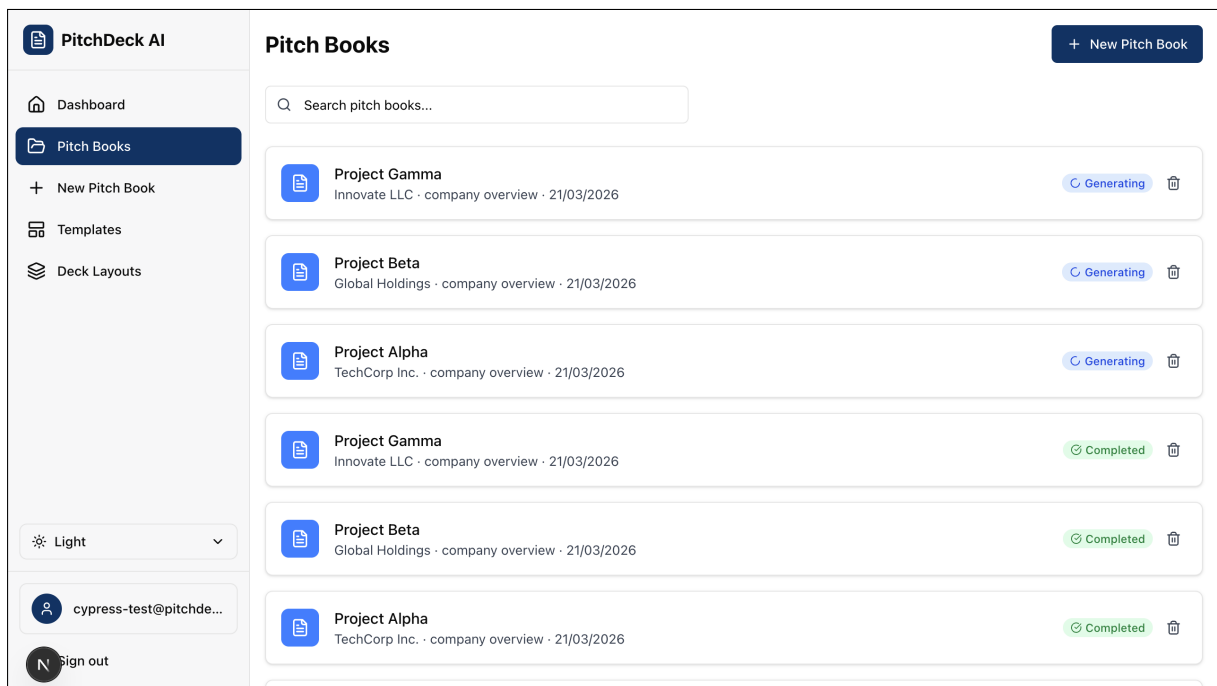


Figure 30: Pitch books list with company name, type, date, and generation status. Includes search filtering.

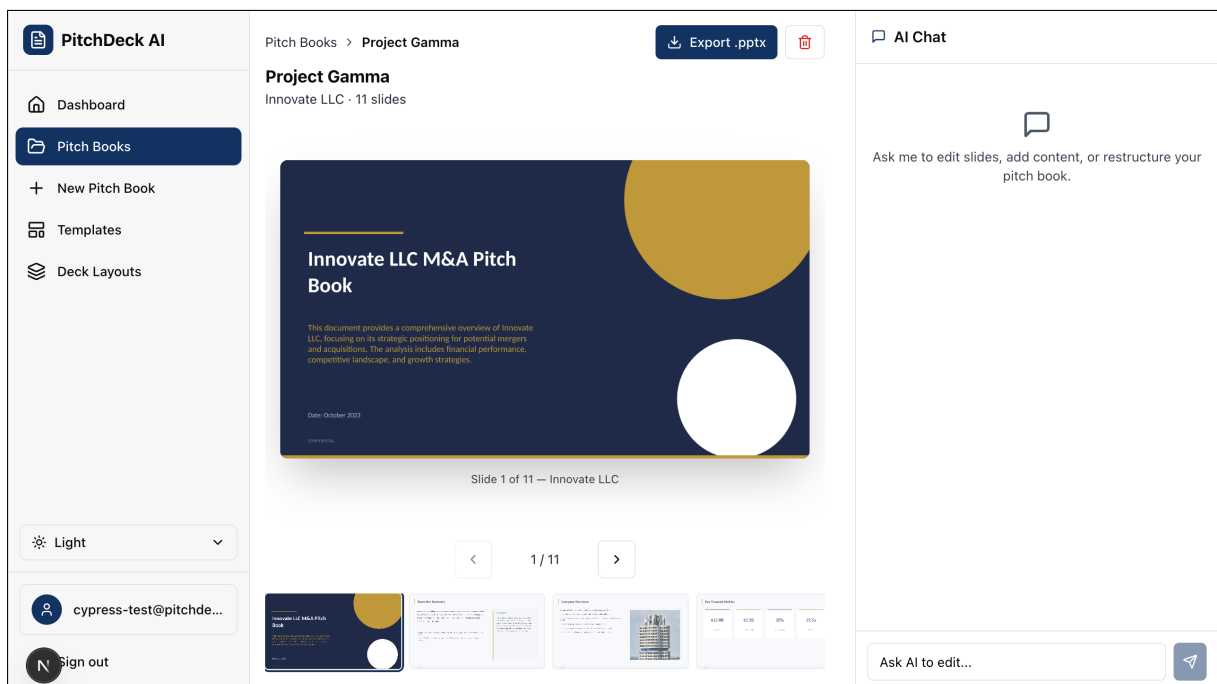


Figure 31: Pitch book viewer with full-size slide, thumbnail strip, and AI chat panel for requesting edits.

Pitch Book Creation

PitchDeck AI

Dashboard

Pitch Books

New Pitch Book

Templates

Deck Layouts

Light

cypress-test@pitchde...

Sign out

New Pitch Book

Fill in the details and let AI generate your pitch book.

Basic Information

Company and presentation details

Pitch Book Title *

Project Delta Generation

Company Name *

Stark Industries

Ticker Symbol

Search ticker or company name...

Presentation Details

Select the type of presentation and transaction context

Pitch Book Type *

Company Overview

Transaction Type

Mergers & Acquisitions

Business overview, financials, and market position

Design Style

Modern

Color Theme

Navy & Gold

Figure 32: New pitch book form with company details, pitch book type, transaction context, and design options.

Templates

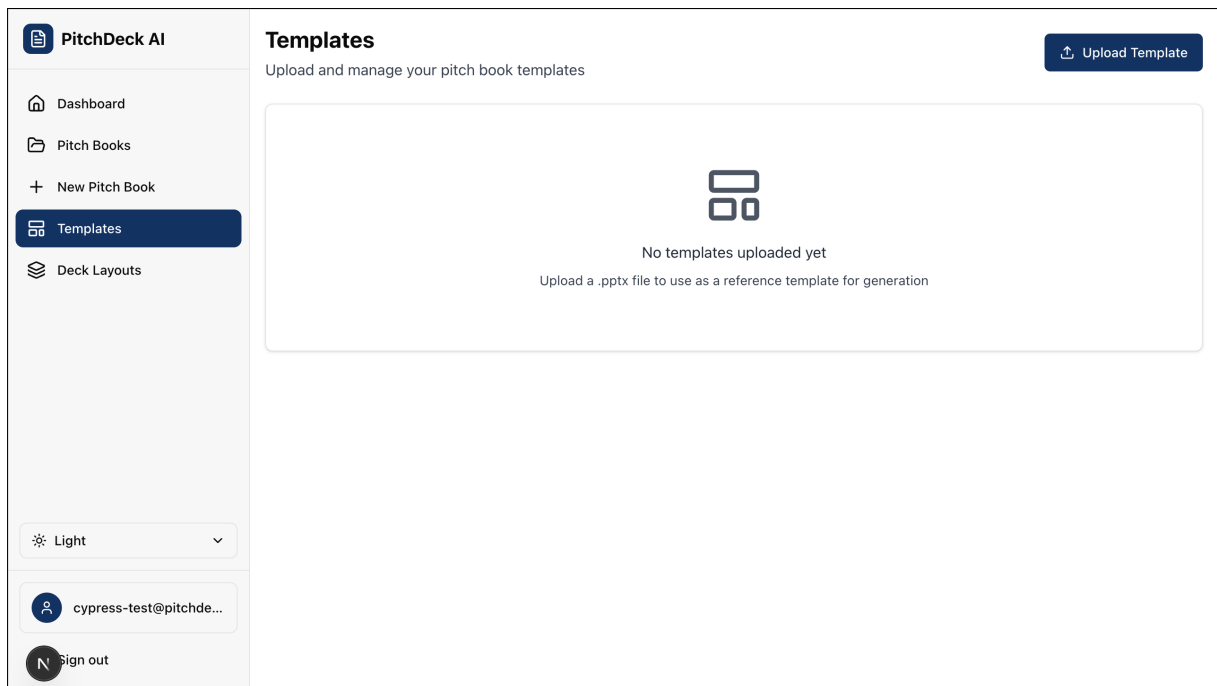


Figure 33: Templates page (empty state) with upload prompt for .pptx reference templates.

Deck Layouts

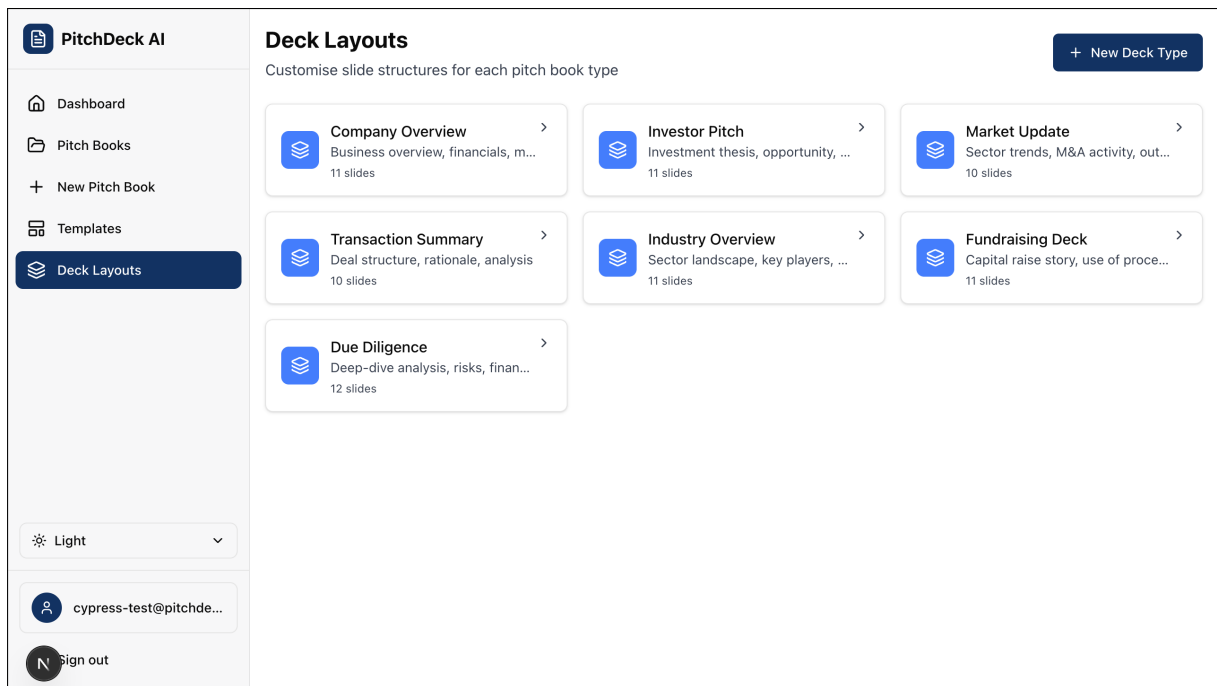


Figure 34: Deck layouts overview with seven pre-configured deck types, each showing a description and slide count.

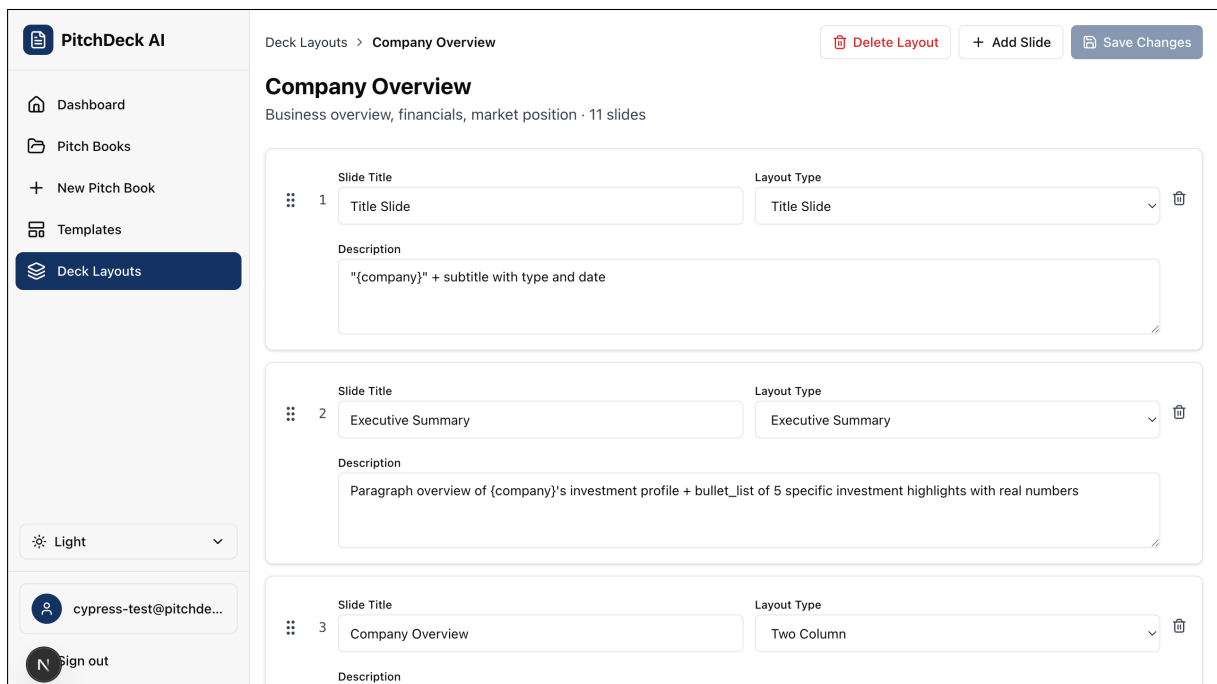


Figure 35: Deck layout editor for Company Overview. Each slide has a title, layout type, and content description. Slides can be reordered, added, or deleted.

Settings

 PitchDeck AI

Dashboard

Pitch Books

+ New Pitch Book

Templates

Deck Layouts

Light

 cypress-test@pitchde...

 Sign out

Settings

Profile

Your account details



cypress-test@pitchdeck.test

Member since 16/03/2026

Change Password

New Password

Min. 8 characters

Update Password

Danger Zone

Sign Out

Figure 36: Settings page with profile details, password change, and sign-out.

Appendix F: PowerPoint Add-in Screenshots

Screenshots of the PowerPoint taskpane add-in running inside Microsoft PowerPoint on macOS. The add-in connects to the same backend as the web application.

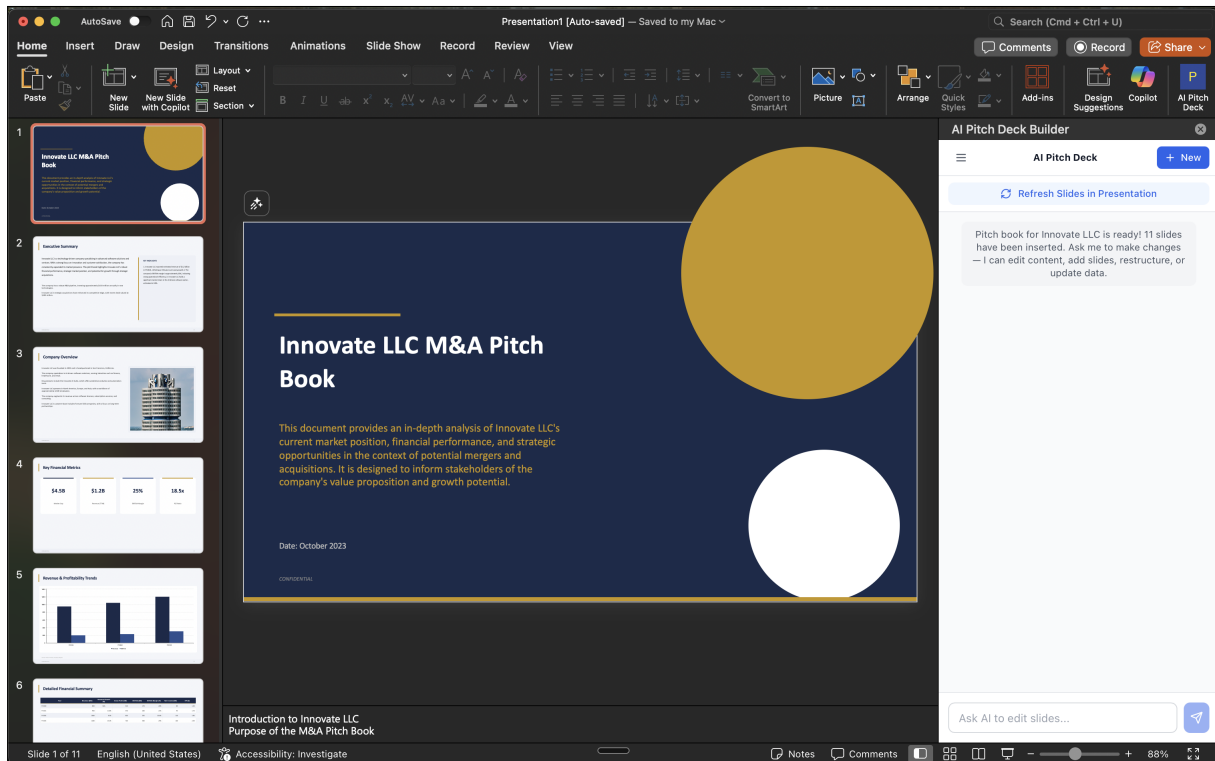


Figure 37: Add-in after generation. 11 slides have been inserted into the presentation. The taskpane shows a completion message and an AI chat input for requesting edits.

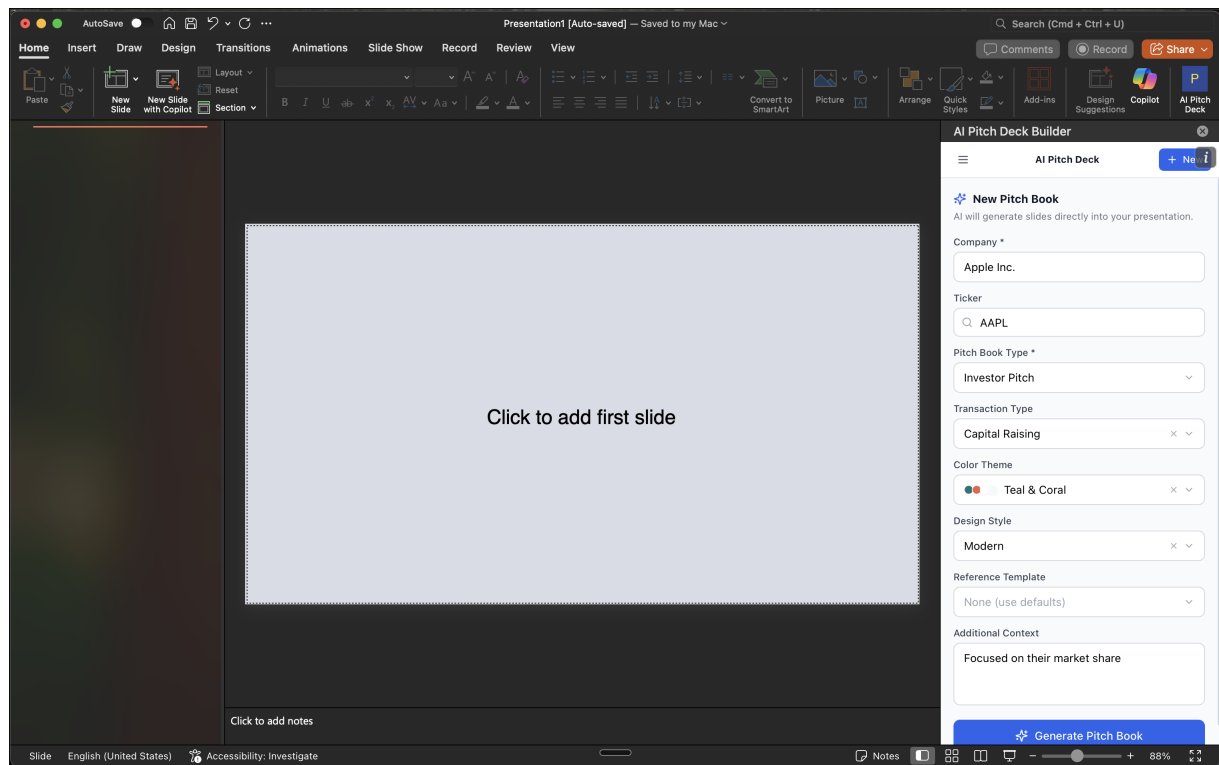


Figure 38: New pitch book form in the add-in. The user enters company name, ticker, pitch book type, transaction type, colour theme, design style, and optional context before generating.

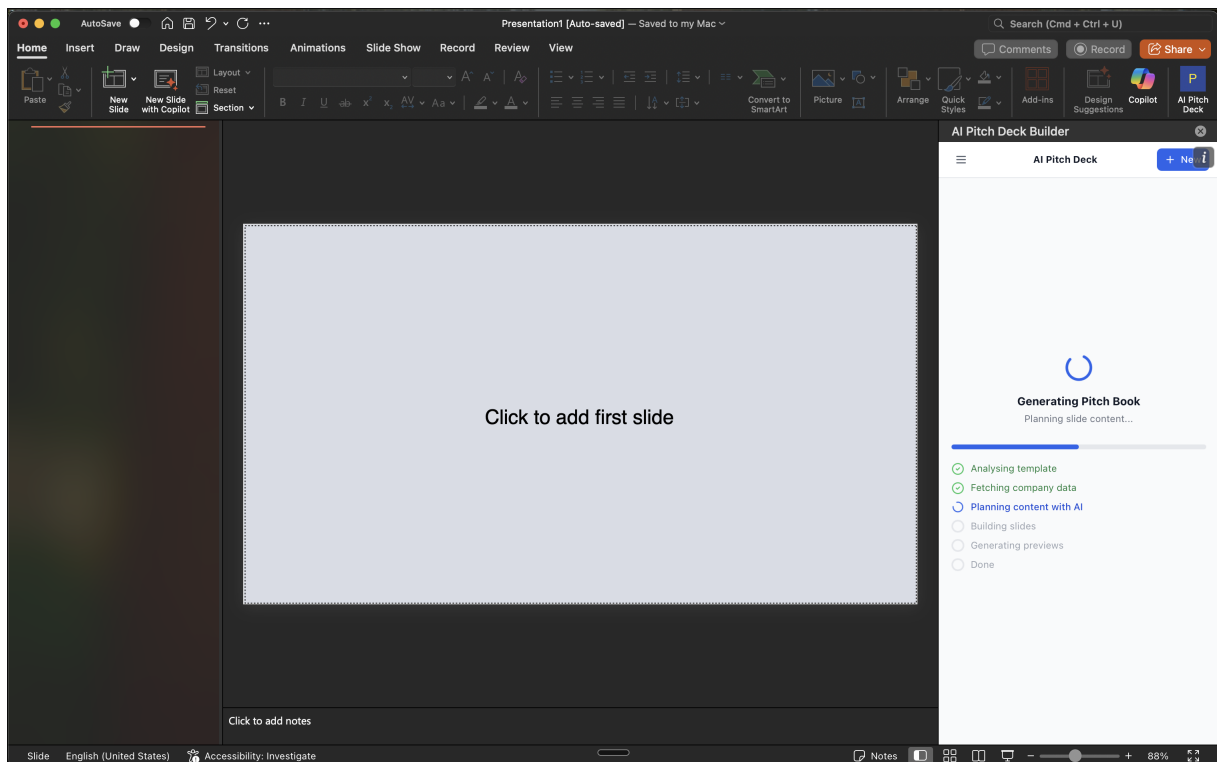


Figure 39: Generation in progress. The taskpane shows a step-by-step progress indicator: analysing template, fetching company data, planning content with AI, building slides, and generating previews.

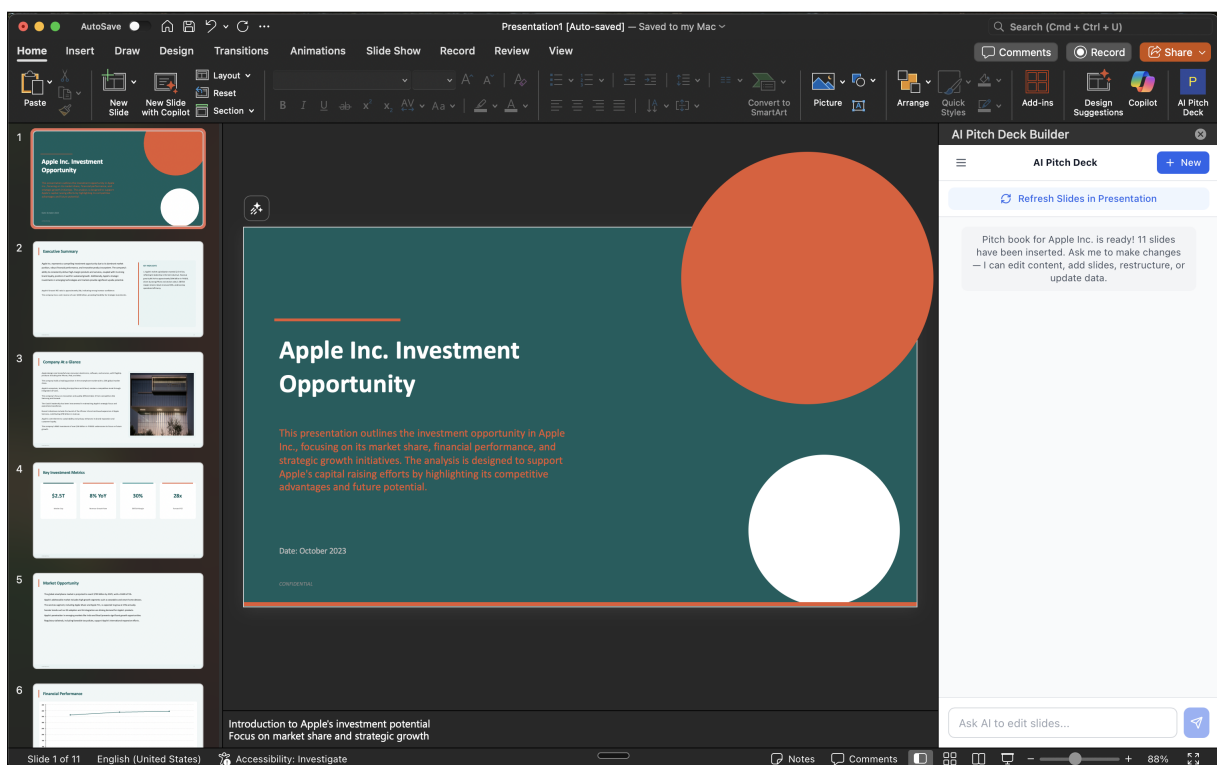


Figure 40: Completed generation for Apple Inc. The teal and coral colour theme was applied automatically. Slides appear in the left panel and the AI chat is ready for edit requests.

Appendix G: Binary Risk Analysis

This appendix shows the full Binary Risk Analysis (BRA) workings for the ten risks listed in Section 2. Sapiro (2011) designed BRA to replace subjective likelihood and impact ratings with a repeatable process. Each risk is scored against ten yes/no questions, and the answers pass through a fixed set of lookup matrices to produce a Low, Medium, or High classification.

Risk Register

Table 33 repeats the risk descriptions from the main report for reference.

ID	Risk	Description
R1	Pitch Book Data Breach	A flaw in Supabase row-level security policies could expose one user's pitch books to another. Pitch books often contain deal terms, valuations, and financial projections that are highly confidential.
R2	Financial Figure Hallucination	GPT-4o could insert fabricated market caps, revenue figures, or growth rates into slides. An analyst presenting incorrect numbers could damage the firm's credibility.
R3	Malicious Template Upload	The template analyser parses .pptx files (ZIP archives with XML) using JSZip and xml2js. A crafted file could attempt zip bomb decompression, path traversal, or oversized XML payloads.
R4	Yahoo Finance End-point Change	yahoo-finance2 scrapes unofficial endpoints. Yahoo has changed these before without notice, which would break financial data retrieval mid-generation.
R5	OpenAI Credential Exposure	The Express backend stores the OpenAI API key on Render. If the key leaks through logs or a deployment compromise, an attacker could run up API costs.
R6	Content Plan Schema Failure	GPT-4o occasionally returns malformed JSON or nests output in unexpected wrappers. If Zod validation and unwrapping both fail, generation stops entirely.
R7	Slide Preview Exposure	LibreOffice converts PPTX files to PNG previews in Supabase Storage. If the bucket defaults to public, anyone with a URL could view confidential slides.
R8	Platform Dependency Outage	Generation chains Vercel, Render, Supabase, and OpenAI. A failure at any point blocks the pipeline with no self-hosted fallback.
R9	Scope Creep Beyond Timeline	The fixed 13-week timeline means unplanned features risk delaying core deliverables.
R10	Add-in Session Breakage	The Office.js iframe uses SameSite=None; Secure cookies for authentication. A browser or Office update could silently break login.

Table 33: Risk register for Binary Risk Analysis.

BRA Worksheet

Each risk was scored against the ten standard BRA questions (listed at the end of this appendix). Questions 1 through 6 feed into the likelihood calculation and questions 7 through 10 feed into the impact calculation. Table 34 shows the raw answers and the computed results.

	Likelihood						Impact				Result		
ID	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	L	I	O
R1	Y	N	N	N	N	N	Y	Y	Y	Y	Low	High	Med
R2	Y	Y	N	N	N	Y	Y	N	Y	N	Med	Med	Med
R3	Y	N	N	N	N	N	Y	N	N	N	Low	Low	Low
R4	Y	Y	N	Y	Y	Y	Y	N	Y	N	High	Med	High
R5	Y	N	N	N	N	N	Y	N	N	N	Low	Low	Low
R6	Y	Y	N	N	N	Y	Y	N	Y	N	Med	Med	Med
R7	Y	N	N	N	N	N	Y	Y	Y	N	Low	High	Med
R8	Y	Y	N	N	Y	Y	Y	Y	Y	N	High	High	High
R9	Y	Y	N	N	Y	N	Y	N	Y	N	Med	Med	Med
R10	Y	Y	N	Y	N	N	Y	N	N	N	Med	Low	Med

Table 34: BRA worksheet. L = Likelihood, I = Impact, O = Overall. Med = Medium.

How the Matrices Work

The ten binary answers do not simply get counted. Instead, they pass through a chain of lookup matrices that combine pairs of inputs into intermediate classifications, which then combine again until a final rating emerges.

Pair-to-pair mappings (2×2)

When two binary answers are paired, the result follows a simple rule:

- Both No → Low
- One Yes, one No (either order) → Medium
- Both Yes → High

Combining intermediate results (3×3)

When two intermediate classifications (each Low, Medium, or High) need to be combined, the following matrix applies:

	Low	Medium	High
Low	Low	Medium	High
Medium	Medium	Medium	High
High	Medium	High	High

Table 35: 3×3 combination matrix used at each stage.

Likelihood chain (Q1–Q6)

The six likelihood questions feed through five matrices in sequence:

1. **Threat Scope** (2×2): Q1 + Q2. Classifies the size of the potential attacker pool as small, medium, or large.
2. **Protection Level** (2×2): Q3 + Q4. Classifies how complete the existing defences are.
3. **Attack Effectiveness** (3×3): Combines Threat Scope with Protection Level to estimate how effective an attack attempt would be.
4. **Occurrence Frequency** (2×2): Q5 + Q6. Classifies how often the vulnerability is exposed.
5. **Overall Likelihood** (3×3): Combines Attack Effectiveness with Occurrence Frequency to produce the final likelihood rating.

Impact chain (Q7–Q10)

The four impact questions feed through three matrices:

1. **Harm** (2×2): Q7 + Q8. Classifies the severity of consequences as limited, material, or damaging.
2. **Asset Value** (2×2): Q9 + Q10. Classifies how critical the affected asset is to the business.
3. **Overall Impact** (3×3): Combines Harm with Asset Value to produce the final impact rating.

The overall severity for each risk is then produced by one final 3×3 combination of likelihood and impact.

BRA Questions

The ten questions used to evaluate each risk are reproduced below. Questions 1–6 address likelihood and questions 7–10 address impact.

1. Can the threat event be carried out by someone with ordinary technical skills?
2. Can the threat event be carried out without significant time, money, or specialist equipment?
3. Is the affected asset currently unprotected against this threat?
4. Are there known gaps or weaknesses in the existing defences?
5. Is the vulnerability present at all times (not just under specific conditions)?
6. Can the threat event happen without any special pre-conditions being met first?
7. Will there be negative consequences felt within the organisation?

8. Will there be negative consequences felt outside the organisation (e.g. clients, regulators)?
9. Does the affected asset contribute significant business value?
10. Would the cost of repairing or replacing the affected asset be significant?